

Recall — data handling, and the project topic

Tally, frequency, the three averages and the range — and the question the project will answer.

LESSON 157

1 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 6 COLLECTING DATA

LESSON CLOCK

8'

12'

12'

8'

Collecting and recording	8 min
The three averages and the range	12 min
The mean from a frequency table	12 min
The project topic	8 min

LEARNING OBJECTIVES

- Record data in a tally and turn it into a frequency table.
- Find the mean, the median, the mode and the range of a small set.
- Find the mean from a frequency table without listing every value.
- Frame a question that a survey can actually answer.

TEXTBOOK

National	pp. 449–452
Cambridge	Stage 7, pp. 58–62
Workbook	Exercise 6.1

01

Explanation

Collecting and recording

Data is either a category — a favourite sport, a colour — or a number. Numbers that can only be counted, such as marks out of ten, are **discrete**; those that are measured, such as heights, are **continuous**.

MARK	TALLY (A CROSSED GROUP IS FIVE)	FREQUENCY
4		1
5		2
6		3
7	/	5
8		4
9		3
10		2

THE FREQUENCIES MUST ADD TO THE NUMBER OF PEOPLE ASKED

Here $1 + 2 + 3 + 5 + 4 + 3 + 2 = 20$. That single check catches most recording errors before any calculation begins.

The three averages and the range

MEASURE	WHAT IT IS	FOR 2, 5, 5, 6, 12
mean	the total shared out equally	$30 \div 5 = 6$
median	the middle value, in order	5
mode	the value that occurs most	5
range	largest minus smallest	$12 - 2 = 10$

THE RANGE IS NOT AN AVERAGE

It says how spread out the data is, not where its centre lies. Two classes can share a mean of 6 and have ranges of 2 and 10 — very different classes.

For an even number of values the median is halfway between the two middle ones.
Order the data first, every time; an unordered list gives the wrong median more often than not.

The mean from a frequency table

Twenty marks need not be written out. Multiply each mark by its frequency, add, and divide by the total frequency.

MARK X	FREQUENCY F	$X \cdot F$
4	1	4
5	2	10
6	3	18
7	5	35
8	4	32
9	3	27
10	2	20
total	20	146

$$\text{mean} = \frac{146}{20} = 7.3$$

MODE, MEDIAN AND RANGE COME OFF THE SAME TABLE

The mode is the mark with the largest frequency — 7. The median is between the 10th and 11th values, both of which are 7. The range is $10 - 4 = 6$.

The project topic

THE PROJECT, ANNOUNCED TODAY AND PRESENTED IN LESSON 203

Which sport do the pupils of our school prefer? Ask at least 60 pupils, record the answers in a tally, and present the result as a pie chart with the angle of every sector shown.

STAGE	LESSONS	WHAT IS PRODUCED
plan the survey	158–159	the question and the sample
collect and tally	at home	a frequency table
draw the chart	160–163	a pie chart with angles
interpret it	164–166	three written conclusions
present	203–204	a poster and four minutes of talk

A GOOD QUESTION HAS A SMALL, FIXED LIST OF ANSWERS

“Which sport do you prefer?” with six options can be tallied. “What do you think about sport?” cannot. Deciding the options before asking anyone is half the work of a survey.

02

Worked examples

EXAMPLE 1 Find the mean, median, mode and range of 2, 5, 5, 6, 12.

1

Total 30, so the mean is $30 \div 5 = 6$.

2

In order, the middle value is 5.
Already in order ✓

3

The mode is 5; the range is $12 - 2 = 10$.
Three different “middles”.

ANSWER

Mean 6, median 5, mode 5, range 10

EXAMPLE 2 Find the mean of the twenty marks in the frequency table.

① Multiply each mark by its frequency: 4, 10, 18, 35, 32, 27, 20.

② Total 146 over 20 pupils.

③ $146 \div 20 = 7.3$.
Between 4 and 10 ✓

ANSWER

7.3

EXAMPLE 3 From the same table, find the median and the mode.

① With 20 values the median is between the 10th and 11th.

② Cumulative frequencies: 1, 3, 6, 11 — both fall on 7.

③ The mode is 7, with frequency 5.
Median and mode agree here.

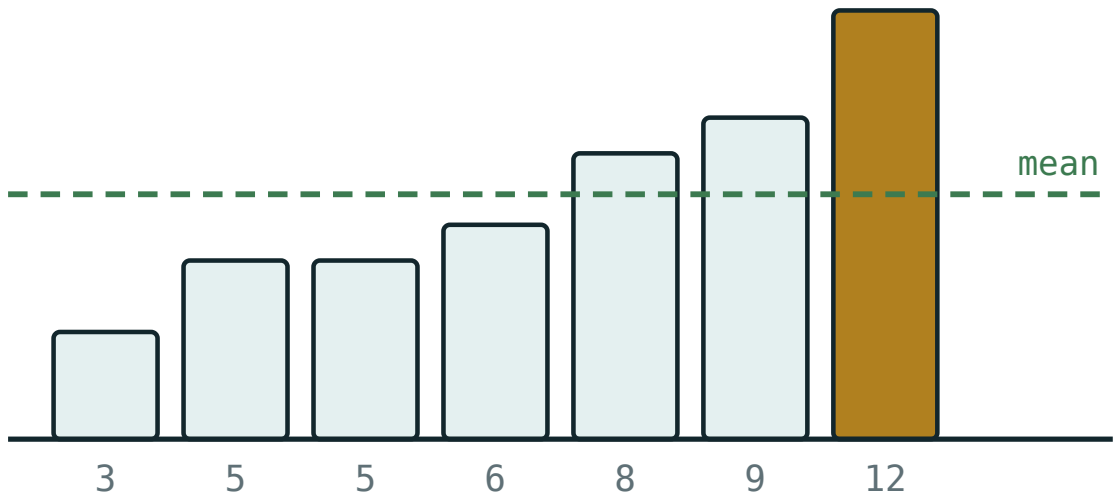
ANSWER

Median 7, mode 7

03 Interactive model

Drag the largest value far to the right in the model: the mean chases it, the median does not. That is the whole argument for the median in one movement.

Mean, median, mode and range



values 3, 5, 5, 6, 8, 9, 12

mean 6.86

median 6

mode 5

range 9

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Data	Ma'lumot	Данные
Tally	Sanoq belgisi	Подсчёт чёрточками
Frequency	Chastota	Частота
Frequency table	Chastotalar jadvali	Таблица частот
Mean	O'rtacha arifmetik	Среднее арифметическое
Median	Mediana	Медиана
Mode	Moda	Мода
Range	Farq (kenglik)	Размах

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The mode is:

the middle value

the most common value

the total $\div$ the count

the spread

The median of 2, 5, 5, 6, 12 is:

5

6

5.5

10

The range of that set is:

5

6

10

12

The mean of the twenty marks is:

7

7.3

146

20

Before finding a median you must:

add the values

order the values

count the modes

draw a chart

Heights of pupils are:

categorical

discrete

continuous

a tally

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The mode of 2, 5, 5, 6, 12

ANSWER 5

2. The median of 2, 5, 5, 6, 12

ANSWER 5

3. The mean of 2, 5, 5, 6, 12

ANSWER 6

4. The range of 2, 5, 5, 6, 12

ANSWER 10

5. The mean of 3, 4, 4, 5, 9

ANSWER 5

6. The mode of the twenty marks

ANSWER 7

7. The number of pupils in the table

ANSWER 20

Medium · the standard question

7 PROBLEMS

1. The median of the twenty marks

ANSWER 7

2. The mean of the twenty marks

ANSWER 7.3

3. The range of the twenty marks

ANSWER 6

4. The total of the twenty marks

ANSWER 146

5. The frequency of the mark 8

ANSWER 4

6. How many scored 8 or more

ANSWER 9

7. How many scored below 7

ANSWER 6

Hard · stretch and reason

7 PROBLEMS

1. A tally of four groups of five and two more

ANSWER 22

2. Which average is least affected by one very large value?

ANSWER The median

3. A 21st pupil scores 5: the new mean, to 1 d.p.

ANSWER 7.2

4. If every mark rose by 1, the new mean

ANSWER 8.3

5. ...and the new range

ANSWER 6, unchanged

6. Which average must always be one of the data values?

ANSWER The mode

7. Data that is measured rather than counted is called

ANSWER Continuous

07 Homework

Homework — 5 tasks

Bring the frequency table of your own survey to the next lesson; the project depends on it.

1. Find the mean, median, mode and range of 3, 7, 7, 8, 10, 13.
2. A frequency table has marks 5, 6, 7, 8 with frequencies 2, 5, 8, 5. Find the mean.
3. Find the median and the mode of that table.
4. Write your survey question with a list of at most six possible answers.
5. Explain in one sentence why the range is not an average.

Conducting an investigation; taking a sample

From a question to a sample that can honestly answer it. [Cambridge insert]

LESSON 158–159

2 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 6.1–6.2 INVESTIGATIONS

LESSON CLOCK

20'

20'

25'

15'

The statistical cycle	20 min
Asking a good question	20 min
Choosing a sample	25 min
Sources of bias	15 min

LEARNING OBJECTIVES

- Name the five stages of a statistical investigation.
- Write a survey question with a small, fixed list of answers.
- Choose a sample that represents the population it is drawn from.
- Recognise the four common sources of bias and remove them.

TEXTBOOK

National	pp. 453–457
Cambridge	Stage 7, pp. 58–66
Workbook	Exercise 6.1–6.2

01

Explanation

The statistical cycle

Every investigation runs through the same five stages, in the same order. Skipping one is what makes a survey worthless.

STAGE	THE WORK	WHAT IT PRODUCES
1 — plan	decide the question and the answers	a survey question
2 — sample	decide who is asked, and how many	a list of people
3 — collect	ask, and record on a sheet	a tally
4 — process	frequencies, averages, a chart	a table and a diagram
5 — interpret	write what the data says	conclusions, with limits

THE QUESTION COMES FIRST, ALWAYS

Data collected before the question is decided almost never answers it. Five minutes of planning saves an afternoon of asking the wrong people the wrong thing.

Asking a good question

POOR QUESTION	WHAT IS WRONG	BETTER
What do you think about sport?	the answers cannot be tallied	Which of these six sports do you prefer?
Don't you agree football is best?	it leads the answer	Which sport do you prefer?
Do you do a lot of sport?	"a lot" means different things	How many hours of sport a week?
How old are you and what sport?	two questions in one	ask them separately

OVERLAPPING OPTIONS ARE AS BAD AS MISSING ONES

Age boxes of 10–12, 12–14 leave a twelve-year-old with two homes and no rule for choosing. Write 10–11, 12–13, 14–15 instead — no gaps, no overlaps.

Choosing a sample

The **population** is everyone the question is about; the **sample** is the part actually asked. A sample of 60 from a school of 600 is one pupil in ten.

GRADE	PUPILS	IN A 1-IN-10 SAMPLE
6	120	12
7	110	11
8	130	13
9	120	12
10	60	6
11	60	6
total	600	60

GIVE EVERY PART OF THE POPULATION ITS SHARE

Each grade contributes a tenth of itself, so the sample looks like the school. Taking all 60 from Grade 6 would give a sample of 60 that answers a different question entirely.

Two honest ways to choose the individuals: draw names from a hat (**random**), or take every tenth name from an alphabetical list (**systematic**). Both are decided before anyone is asked.

Sources of bias

WHAT WAS DONE	WHY THE RESULT IS BIASED	THE FIX
only the football team was asked	they already prefer football	sample across the school
only pupils in the canteen at lunch	it misses those who bring food	ask from a class list
only five pupils were asked	too few for any conclusion	at least 60
the question was “isn’t football best?”	the answer is suggested	list the sports neutrally

A LARGE BIASED SAMPLE IS STILL BIASED

Asking 500 people at a football match which sport they prefer is worse than asking 60 pupils chosen properly. Size does not repair a bad sample — only the choosing does.

02 Worked examples

EXAMPLE 1 Plan a 1-in-10 sample of a school of 600 pupils spread over six grades.

- 1 $600 \div 10 = 60$ pupils in the sample.
- 2 Each grade gives a tenth of itself: 12, 11, 13, 12, 6, 6.
- 3 Those add to 60.
The sample now looks like the school ✓

ANSWER

60 pupils, 12, 11, 13, 12, 6 and 6 by grade

EXAMPLE 2 A pupil surveys the football team about the school's favourite sport. Say what is wrong and fix it.

- 1 The sample is not representative — every member already prefers football.
- 2 The conclusion would be true of the team, not of the school.
- 3 Take every tenth name from the whole school list instead.
Same effort, honest answer.

ANSWER

A biased sample; sample across the whole school

EXAMPLE 3 Improve the question “Do you do a lot of sport?”

- ① “A lot” is not defined, so two pupils answer differently for the same behaviour.
- ② Ask for a number instead.
- ③ “How many hours of sport did you do last week?”
Now the answers can be averaged.

ANSWER

“How many hours of sport did you do last week?”

03 Interactive model

Ask the class the leading question first and count the hands; then ask the neutral one and count again. The gap between the two counts is the lesson.

Planning an honest survey

The whole group being studied is the:

sample

population

tally

frequency

A 1-in-10 sample of 600 pupils has:

6

10

60

600

A grade of 130 contributes:

10

13

30

60

Asking only the football team gives a sample that is:

random

biased

systematic

too large

“Don’t you agree football is best?” is:

neutral

leading

too long

fine

Age boxes *10–12* and *12–14* are wrong because they:

overlap

have gaps

are too wide

are too narrow

Taking every tenth name from a list is a sample that is:

biased

systematic

too small

leading

Asking 500 people at a football match is:

fine, it is large

still biased

random

systematic

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Investigation	Tadqiqot	Исследование
Population	Bosh to'plam	Генеральная совокупность
Sample	Tanlanma	Выборка
Representative	Vakillik qiluvchi	Репрезентативный
Bias	Yon bosish	Смещение
Random	Tasodifiy	Случайный
Leading question	Yo'naltiruvchi savol	Наводящий вопрос
Recording sheet	Qayd varaqasi	Лист записи

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The first stage of an investigation is:

collecting

planning the question

drawing a chart

sampling

A sample should be:

as small as possible

representative of the population

taken from one class

chosen after the data

A good survey question has:

open answers

a small fixed list of answers

two questions in one

no options

Bias means the sample:

is too large

favours part of the population

is random

has no answers

Options must have:

gaps

overlaps

neither gaps nor overlaps

both

The last stage of the cycle is:

collecting

tallying

interpreting and reporting

planning

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The whole group being studied is called

ANSWER The population

2. The part actually asked is called

ANSWER The sample

3. 60 out of 600 as a fraction

ANSWER $\frac{1}{10}$

4. ...as a percentage

ANSWER 10%

5. A sample that favours one group is

ANSWER Biased

6. The first stage of the statistical cycle

ANSWER Planning the question

7. A question with a fixed list of answers can be

ANSWER Tallied

Medium · the standard question

7 PROBLEMS

1. A 1-in-10 sample of 600 pupils

ANSWER 60

2. Grade 8 has 130 pupils: its share of that sample

ANSWER 13

3. Grades of 120, 110, 130, 120, 60, 60: the total

ANSWER 600

4. A school of 800 with a 1-in-20 sample

ANSWER 40

5. Why ask 60 rather than 6?

ANSWER A very small sample is unreliable

6. Asking only the football team is

ANSWER Biased

7. "Don't you agree football is best?" is

ANSWER A leading question

Hard · stretch and reason

7 PROBLEMS

1. A school of 750 with a 1-in-25 sample

ANSWER 30 pupils

2. 300 boys and 300 girls in a sample of 60

ANSWER 30 and 30

3. 400 boys and 200 girls in a sample of 60

ANSWER 40 and 20

4. Grades of 120 and 60 in a 1-in-10 sample

ANSWER 12 and 6

5. Every tenth name from a list of 600 gives

ANSWER 60 — a systematic sample

6. One reason a lunchtime canteen survey is biased

ANSWER It misses those who bring food from home

7. The last stage of the cycle

ANSWER Interpreting and reporting

07 Homework

Homework — 5 tasks

Bring the finished plan to the next lesson: the question, the options, the sample and the sheet.

1. Write your survey question with at most six options, with no gaps and no overlaps.
2. A school has 540 pupils. Plan a 1-in-10 sample and say how many come from a grade of 90.
3. Explain in one sentence why asking only your own class would bias the result.
4. Rewrite the question “Do you watch a lot of television?” so that it can be averaged.
5. Draw the recording sheet you will use, with a row for every option and a column for the tally.

Representing data on a pie chart

The whole circle is the whole survey — every frequency becomes an angle.

LESSON 160–163

4 × 40 MIN

MATEMATIKA 6, §32

S7 16 INTERPRETING RESULTS

LESSON CLOCK

30'

35'

35'

35'

15'

10'

The whole is 360°	30 min
From frequency to angle	35 min
Drawing the chart	35 min
Percentages and pie charts	35 min
When the numbers are awkward	15 min
Homework	10 min

LEARNING OBJECTIVES

- Explain why the sectors of a pie chart must total 360° .
- Turn a frequency into a sector angle.
- Draw a pie chart accurately with a protractor and label it.
- Turn a percentage into an angle, and check the total.

TEXTBOOK

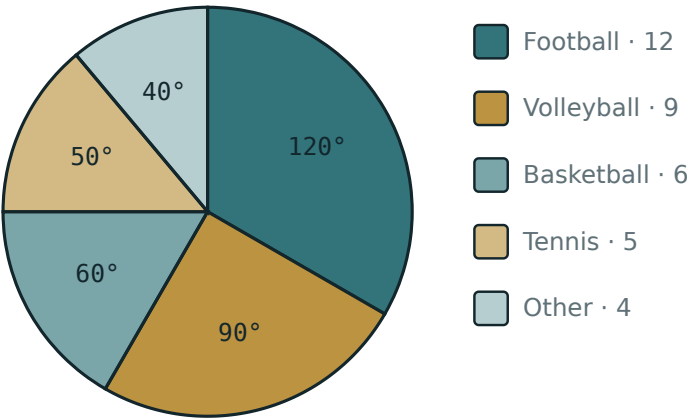
National	pp. 458–469
Cambridge	Stage 7, pp. 174–180
Workbook	Exercise 16.1

01

Explanation

The whole is 360°

A pie chart shows how a whole is divided. The whole circle — 360° — stands for everyone who was asked, and each sector takes the share of 360° that its group deserves.



$$\text{angle} = \frac{\text{frequency}}{\text{total}} \times 360^\circ$$

FIND WHAT ONE PERSON IS WORTH, ONCE

With 36 pupils, $360 \div 36 = 10^\circ$ for each. Every angle is then the frequency times 10 — no fractions at all. With 60 pupils it is 6° each; with 45, 8°.

From frequency to angle

SPORT	PUPILS	WORKING	ANGLE
football	12	$12 \cdot 10$	120°
volleyball	9	$9 \cdot 10$	90°
basketball	6	$6 \cdot 10$	60°
tennis	5	$5 \cdot 10$	50°
other	4	$4 \cdot 10$	40°
total	36	—	360°

IF THE ANGLES DO NOT ADD TO 360°, SOMETHING IS WRONG

Add the last column before drawing anything. A total of 355° or 365° means an arithmetic slip, and it is far easier to find it in the table than in the drawing.

Drawing the chart

STEP	WHAT TO DO	WHY
1	draw a circle and one radius, straight up	a fixed starting line
2	measure the first angle from that radius	the protractor needs a base
3	measure each next angle from the line just drawn	the sectors join up
4	label every sector, or add a key	a chart without labels says nothing
5	write the total and the title	the reader needs the 36

DRAW THE LARGEST SECTOR FIRST

Small errors then accumulate into the smallest sector, where they matter least — and the last sector is what is left over, which is a free check on all the others.

Percentages and pie charts

A percentage is already a share of the whole, so it converts even faster: 100% is 360°, so 1% is 3.6°.

$\text{angle} = \text{percentage} \times 3.6^\circ$

PERCENTAGE	WORKING	ANGLE
50%	$50 \cdot 3.6$	180°
25%	$25 \cdot 3.6$	90°
40%	$40 \cdot 3.6$	144°
12.5%	$12.5 \cdot 3.6$	45°
10%	$10 \cdot 3.6$	36°

THREE LANDMARKS WORTH KNOWING BY HEART

A half is 180° , a quarter 90° , a tenth 36° . Any sector can then be judged by eye before the protractor confirms it.

When the numbers are awkward

Not every total divides 360 exactly. With 7 people each is worth $360 \div 7 = 51.4^\circ$ to one decimal place, and seven of those come to 359.8° .

TOTAL ASKED	ONE PERSON IS WORTH	EXACT?
36	10°	yes
45	8°	yes
50	7.2°	yes
7	51.4°	no — rounded
11	32.7°	no — rounded

ROUND THE ANGLES, THEN GIVE THE LEFTOVER TO THE LARGEST SECTOR

A pie chart must close. If the rounded angles come to 359.8° , add the missing 0.2° to the biggest sector, where it cannot be seen — and say in the working that you did.

02 Worked examples

EXAMPLE 1 Thirty-six pupils name a favourite sport: 12, 9, 6, 5, 4. Find the five sector angles.

① $360 \div 36 = 10^\circ$ for each pupil.

② Multiply each frequency by 10.

③ $120^\circ, 90^\circ, 60^\circ, 50^\circ, 40^\circ$.

They total 360° ✓

ANSWER

$120^\circ, 90^\circ, 60^\circ, 50^\circ$ and 40°

EXAMPLE 2 Of 60 pupils, 9 prefer tennis. Find that sector angle.

① $360 \div 60 = 6^\circ$ for each pupil.

② $9 \cdot 6$.

③ $= 54^\circ$.

Or $\frac{9}{60} \cdot 360$ directly.

ANSWER

54°

EXAMPLE 3 A survey gives 25%, 30%, 20%, 15% and 10%. Find the angles.

① 1% is 3.6° .

② $25 \cdot 3.6 = 90$, $30 \cdot 3.6 = 108$, $20 \cdot 3.6 = 72$.

③ $15 \cdot 3.6 = 54$ and $10 \cdot 3.6 = 36$.

Total 360° ✓

ANSWER

90° , 108° , 72° , 54° and 36°

03 Interactive model

Have the class find “what one person is worth” aloud before any angle is calculated; almost every mistake in this topic is a missed division at that first step.

Frequencies into angles

The sectors of a pie chart total:

100°

180°

360°

the frequency

With 36 pupils, one pupil is worth:

6°

10°

36°

12°

With 60 pupils, one pupil is worth:

6°

10°

60°

3.6°

12 out of 36 gives:

60°

90°

120°

12°

9 out of 60 gives:

45°

54°

60°

90°

25% of a pie chart is:

25°

90°

72°

180°

1% of a pie chart is:

1°

3.6°

36°

0.36°

If the rounded angles total 359.8° you should:

leave a gap

start again

give the extra to the largest sector

divide by 7

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Pie chart	Doiraviy diagramma	Круговая диаграмма
Sector	Sektor	Сектор
Sector angle	Sektor burchagi	Угол сектора
Frequency	Chastota	Частота
Total frequency	Umumiy chastota	Общая частота
Protractor	Transportir	Транспортир
Key (legend)	Izoh	Легенда
Proportion	Ulush	Доля

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The angle of a sector is the frequency divided by the total, times:

With 45 people, one person is worth:

A sector of 180° represents:

A percentage becomes an angle when multiplied by:

3.6

36

360

1.8

Before drawing you should check that the angles:

are all equal

total 360°

are whole numbers

are under 90°

A pie chart without labels or a key:

is still complete

says nothing

is more accurate

needs no total

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The whole circle in degrees

ANSWER 360°

2. 36 pupils: the angle for one pupil

ANSWER 10°

3. 60 pupils: the angle for one pupil

ANSWER 6°

4. 12 out of 36

ANSWER 120°

5. 9 out of 36

ANSWER 90°

6. 6 out of 36

ANSWER 60°

7. The angles of a pie chart add to

ANSWER 360°

Medium · the standard question

7 PROBLEMS

1. 20 out of 60

ANSWER 120°

2. 15 out of 60

ANSWER 90°

3. 9 out of 60

ANSWER 54°

4. 6 out of 60

ANSWER 36°

5. 8 out of 24

ANSWER 120°

6. 3 out of 24

ANSWER 45°

7. 45 people: the angle for one

ANSWER 8°

Hard · stretch and reason

7 PROBLEMS

1. 25% of a pie chart

ANSWER 90°

2. 40%

ANSWER 144°

3. 12.5%

ANSWER 45°

4. 7 out of 40

ANSWER 63°

5. 13 out of 50

ANSWER 93.6°

6. 7 people in total: the angle for one, to 1 d.p.

ANSWER 51.4°

7. Why must the drawn angles be checked against 360° ?

ANSWER Rounding can leave the chart open

07 Homework

Homework — 5 tasks

Every pie chart needs a title, a total, and a label or key on every sector.

1. A survey has 30 pupils. Find the angle representing one pupil.
2. Draw the pie chart for frequencies 10, 8, 6, 4, 2 out of 30, with every angle labelled.
3. Of 200 people, 70 prefer tea. Find that sector angle.
4. Find the angles for the percentages 45%, 30%, 15% and 10%.
5. Draw the pie chart of your own survey data, with the total and a key.

Reading and interpreting data from pie charts

The same rule run backwards — and the questions a pie chart is not able to answer.

LESSON 164–166

3 × 40 MIN

MATEMATIKA 6, §33

S7 16 INTERPRETING RESULTS

LESSON CLOCK

30'

30'

35'

20'

5

From an angle back to a frequency	30 min
Angles as percentages	30 min
Comparing two pie charts	35 min
What a pie chart cannot tell you	20 min
Homework	5 min

LEARNING OBJECTIVES

- Turn a sector angle back into a frequency when the total is known.
- Express a sector as a percentage of the whole.
- Compare two pie charts drawn from different totals.
- State what a pie chart cannot show, and what extra information is needed.

TEXTBOOK

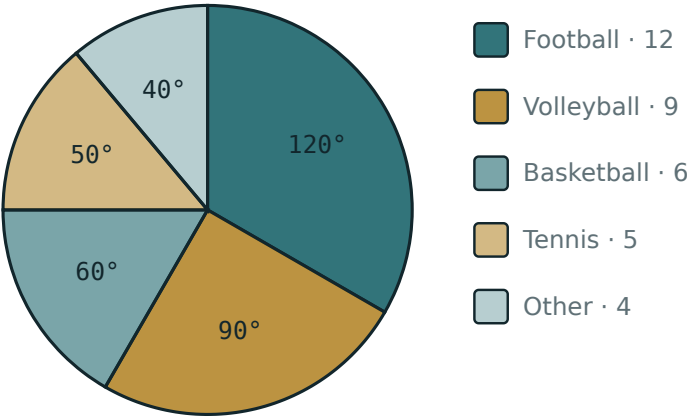
National	pp. 470–479
Cambridge	Stage 7, pp. 178–184
Workbook	Exercise 16.2

01

Explanation

From an angle back to a frequency

Drawing the chart divided 360 by the total. Reading it does the reverse: divide the angle by what one person was worth.



36 pupils · each pupil is 10°

The same chart, now being read rather than drawn

$$\text{frequency} = \frac{\text{angle}}{360^\circ} \times \text{total}$$

TOTAL ASKED	SECTOR	WORKING	FREQUENCY
36	120°	$120 \div 10$	12
60	54°	$54 \div 6$	9
200	90°	$\frac{90}{360} \cdot 200$	50
500	72°	$\frac{72}{360} \cdot 500$	100
240	45°	$\frac{45}{360} \cdot 240$	30

THE TOTAL MUST BE GIVEN SOMEWHERE

A 90° sector is a quarter of *something*. Without the total written on the chart, it is a quarter of an unknown number — and no frequency can be recovered at all.

Angles as percentages

A percentage needs no total: it is the share itself. Divide the angle by 3.6.

$$\text{percentage} = \frac{\text{angle}}{3.6}$$

SECTOR	WORKING	PERCENTAGE	AS A FRACTION
180°	$180 \div 3.6$	50%	$\frac{1}{2}$
144°	$144 \div 3.6$	40%	$\frac{2}{5}$
90°	$90 \div 3.6$	25%	$\frac{1}{4}$
54°	$54 \div 3.6$	15%	$\frac{3}{20}$
36°	$36 \div 3.6$	10%	$\frac{1}{10}$

JUDGE BY EYE, THEN CONFIRM

A sector that looks like a quarter should come out near 25%. If the arithmetic gives 2.5% or 250%, the division went the wrong way round.

Comparing two pie charts

Two charts of the same size can stand for very different numbers of people. The angles compare **shares**; only the totals compare **numbers**.

SCHOOL	PUPILS	FOOTBALL SECTOR	FOOTBALL PUPILS
A	200	90°	50
B	500	90°	125
C	180	120°	60
D	400	90°	100

A LARGER SHARE CAN BE A SMALLER NUMBER

School C gives football a third of its chart and school D only a quarter — yet D has 100 footballers and C has 60. Comparing the angles alone would have reversed the answer.

What a pie chart cannot tell you

QUESTION	CAN THE CHART ANSWER IT?	WHAT IS NEEDED
which group is the largest?	yes	nothing — the biggest sector
what share does a group have?	yes	the angle
how many people is that?	no	the total surveyed
has it changed since last year?	no	a second chart and both totals
what is the mean?	no	the original values, not the shares

THREE SENTENCES MAKE A GOOD INTERPRETATION

The largest group and its share; a comparison between two groups; and one honest limit — usually the sample, its size, or something the chart cannot show.

02 Worked examples

EXAMPLE 1 A pie chart of 60 pupils has a 54° sector for tennis. How many pupils is that?

1 $360 \div 60 = 6^\circ$ for each pupil.

2 $54 \div 6$.

3 = 9 pupils.
Check: $9 \cdot 6 = 54$ ✓

ANSWER

9 pupils

EXAMPLE 2 A survey of 500 people has a 72° sector. Find the number and the percentage.

$$\textcircled{1} \quad \frac{72}{360} = \frac{1}{5}.$$

$$\textcircled{2} \quad \frac{1}{5} \cdot 500 = 100 \text{ people.}$$

$$\textcircled{3} \quad 72 \div 3.6 = 20\%.$$

A fifth ✓

ANSWER

100 people, or 20%

EXAMPLE 3 School A has 200 pupils and school B has 500. Football is 90° in both charts. Compare them.

$\textcircled{1}$ The shares are equal — a quarter each.

$\textcircled{2}$ 50 pupils in A, 125 in B.

$\textcircled{3}$ Same proportion, very different numbers.
The angle alone cannot compare schools.

ANSWER

Equal shares; 50 against 125 pupils

03 Interactive model

Show two charts side by side with the totals covered, ask which school has more footballers, then uncover the totals. The reversal is the point of the lesson.

Reading the chart

A 120° sector of 36 pupils is:

A 54° sector of 60 pupils is:

A 90° sector of 200 people is:

A 144° sector as a percentage is:

14.4%

40%

44%

60%

A 36° sector as a percentage is:

3.6%

10%

36%

20%

Without the total, a chart gives:

the numbers

the shares only

the mean

nothing at all

90° of 400 against 120° of 180:

100 against 60

60 against 100

equal

cannot tell

A pie chart cannot show:

the largest group

a share

the change over time

a comparison of two sectors

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
To interpret	Talqin qilmoq	Интерпретировать
Sector angle	Sektor burchagi	Угол сектора
Proportion	Ulush	Доля
Total surveyed	So'ralganlar soni	Число опрошенных
To compare	Taqqoslamog	Сравнивать
Conclusion	Xulosa	Вывод
Misleading	Chalg'ituvchi	Вводящий в заблуждение
Estimate	Taxminiy baho	Оценка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

To turn a sector into a frequency you need:

the angle only

the angle and the total

the total only

a protractor

A 180° sector is:

a quarter

a half

a third

everything

To turn an angle into a percentage divide by:

3.6

36

360

100

Two charts with equal sectors represent:

equal numbers

equal shares

nothing comparable

equal totals

A larger sector always means:

more people

a larger share

a larger total

a better result

A good interpretation ends with:

a bigger chart

an honest limit

the mean

the angles

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 36 pupils, a sector of 120°

ANSWER 12 pupils

2. 36 pupils, a sector of 90°

ANSWER 9 pupils

3. 60 pupils, a sector of 54°

ANSWER 9 pupils

4. 60 pupils, a sector of 36°

ANSWER 6 pupils

5. A sector of 180° as a fraction

ANSWER $\frac{1}{2}$

6. A sector of 90° as a percentage

ANSWER 25%

7. A sector of 36° as a percentage

ANSWER 10%

Medium · the standard question

7 PROBLEMS

1. 200 people, a sector of 90°

ANSWER 50 people

2. 500 people, a sector of 72°

ANSWER 100 people

3. 240 people, a sector of 45°

ANSWER 30 people

4. A sector of 144° as a percentage

ANSWER 40%

5. A sector of 54° as a percentage

ANSWER 15%

6. 90 people, a sector of 100°

ANSWER 25 people

7. 720 people, a sector of 30°

ANSWER 60 people

Hard · stretch and reason

7 PROBLEMS

1. A sector of 60° as a percentage, to 1 d.p.

ANSWER 16.7%

2. Football at 90° in schools of 200 and 500

ANSWER 50 and 125 pupils

3. 120° of 180 against 90° of 400

ANSWER 60 against 100

4. Which of those two has more footballers?

ANSWER The second, despite the smaller sector

5. What can a pie chart never give without more information?

ANSWER The number of people surveyed

6. A sector estimated at 100° out of 90 pupils

ANSWER 25 pupils

7. Two pie charts can be compared by number only when

ANSWER Both totals are known

07 Homework

Homework — 5 tasks

Write the total beside every chart you read; without it no frequency can be found.

1. A pie chart of 72 pupils has a 100° sector. How many pupils does it represent?
2. The same chart has a 45° sector. How many pupils is that?
3. Express 144° and 54° as percentages of the whole.
4. In school A (300 pupils) football is 120° ; in school B (600) it is 90° . Which school has more footballers?

5. Write two questions that a pie chart on its own cannot answer.

Control work 9

— data handling

Frequency tables, the three averages, and pie charts drawn and read.

LESSON 167

1 × 40 MIN

MATEMATIKA 6, NAZORAT ISHI 9

S7 6 AND 16 REVIEW

LESSON CLOCK

2'

28'

8'

2'

Instructions	2 min
The paper	28 min
Answers and diagnosis	8 min
What comes next	2 min

LEARNING OBJECTIVES

- Complete a frequency table and find its total.
- Find the mean, median, mode and range from a table.
- Turn a frequency into a sector angle and back again.
- Name each lost mark by its type and rewrite the whole solution.

TEXTBOOK

National	pp. 449–479
Cambridge	Stage 7, pp. 58–184
Workbook	Control paper 9

01

Explanation

The paper — 20 marks, 28 minutes

Questions 1 to 4 use this table of marks out of ten for one class.

MARK	5	6	7	8
frequency	3	6	7	4

Q	TASK	MARKS	FROM
1	The number of pupils in the class	2	L157
2	The mode and the range	3	L157
3	The median	3	L157
4	The mean	3	L157
5	A survey of 45 pupils: the sector angle for a group of 10	3	L160–163
6	A pie chart of 200 people has a 63° sector: how many people?	3	L164–166
7	Express 108° as a percentage of the whole	2	L164–166
8	Give one reason a survey of the school football team is biased	1	L158–159

THE ANSWERS

20 pupils; mode 7, range 3; median 7; mean 6.6; 80° ; 35 people; 30%; the team already prefers football.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
the mean divided by the number of marks	$132 \div 4$	$132 \div 20$
the range given as an interval	5 to 8	3
the median taken as the middle mark of the row	6.5	7
the angle taken as the frequency over the total	$10 \div 45$	$10 \cdot 8 = 80^\circ$
the sector read as the number of people	63	35
the percentage division reversed	$3.6 \div 108$	$108 \div 3.6$
the degree sign missing	80	80°

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

What comes next

IF YOU LOST MARKS ON	REVISE
Q1–Q4	frequency tables and the averages, L157
Q5	drawing a pie chart, L160–163
Q6–Q7	reading a pie chart, L164–166
Q8	samples and bias, L158–159

LOOKING FORWARD

Next comes the choice of chart for a given kind of data, then graphs of relationships, probability, the transformations on a grid, and the three-dimensional shapes.

02 Worked examples

EXAMPLE 1 Model answer, Q3: the median of the twenty marks.

- 1 With 20 values the median is between the 10th and 11th.
- 2 Cumulative frequencies: 3, 9, 16, 20.
Both fall in the mark 7.
- 3 The median is 7.

ANSWER

7

EXAMPLE 2 Model answer, Q5: 45 pupils, a group of 10.

① $360 \div 45 = 8^\circ$ for each pupil.
What one person is worth.

② $10 \cdot 8.$

③ $= 80^\circ.$
Under a quarter of the circle ✓

ANSWER 80° **EXAMPLE 3** Model answer, Q6: a 63° sector of 200 people.

① $\frac{63}{360}$ of the survey.

② $\frac{63}{360} \cdot 200.$

③ $= 35$ people.
Under a fifth of 200 ✓

ANSWER

35 people

03 Interactive model

Return Q4 first and ask who divided by four; naming it as “dividing by the number of different marks” fixes it permanently.

The paper in eight questions

Frequencies 3, 6, 7, 4 give a class of:

The mode of that table is:

The range is:

The median is:

6

6.5

7

7.5

The mean is:

6.6

33

7

6.5

With 45 pupils, 10 of them take:

45°

80°

10°

36°

A 63° sector of 200 people is:

63

35

50

20

108° as a percentage is:

10.8%

30%

36%

108%

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Frequency table	Chastotalar jadvali	Таблица частот
Mean	O‘rtacha arifmetik	Среднее арифметическое
Median	Mediana	Медиана
Mode	Moda	Мода
Range	Farq (kenglik)	Размах
Sector angle	Sektor burchagi	Угол сектора
Bias	Yon bosish	Смещение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 is answered by:

adding the marks

adding the frequencies

counting the columns

dividing

Q4 divides the total of the marks by:

4

20

132

8

Q5 first finds:

the percentage

what one pupil is worth

the total angle

the median

Q6 needs:

the angle only

the angle and the total

the total only

the mode

Q7 needs no total because a percentage is:

a number

a share

an angle

a frequency

Q8 is about:

arithmetic

a biased sample

the mean

a chart

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The total of frequencies 3, 6, 7, 4

ANSWER 20

2. The mode of that table

ANSWER 7

3. The range of marks from 5 to 8

ANSWER 3

4. 45 pupils: the angle for one

ANSWER 8°

5. 10 pupils out of 45

ANSWER 80°

6. 108° as a percentage

ANSWER 30%

7. The angles of a pie chart total

ANSWER 360°

Medium · the standard question

7 PROBLEMS

1. The median of that table

ANSWER 7

2. The mean of that table

ANSWER 6.6

3. The total of the twenty marks

ANSWER 132

4. 200 people, a sector of 63°

ANSWER 35 people

5. How many scored 7 or more

ANSWER 11

6. How many scored below 7

ANSWER 9

7. The frequency of the mark 6

ANSWER 6

Hard · stretch and reason

7 PROBLEMS

1. Why is $132 \div 4$ the wrong mean?

ANSWER It divides by the number of different marks

2. 45 pupils, a group of 18: the angle

ANSWER 144°

3. 200 people, a sector of 90°

ANSWER 50 people

4. A 21st pupil scores 8: the new mean, to 2 d.p.

ANSWER 6.67

5. A survey of the football team gives

ANSWER Too large a share for football

6. 90° as a percentage

ANSWER 25%

7. The median of an even number of values is

ANSWER Halfway between the two middle values

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, with the units and the degree signs.

1. Rewrite in full every question on which you lost a mark, naming the slip in the margin.
2. Marks 4, 5, 6, 7 have frequencies 2, 5, 8, 5. Find the mean, median, mode and range.
3. A survey of 90 pupils has a group of 25. Find the sector angle.
4. A pie chart of 300 people has a 126° sector. How many people is that?
5. Express 72° as a percentage and as a fraction of the whole.

Think — choosing a chart

Four kinds of diagram, four kinds of question — and the charts that quietly mislead.

LESSON 168

1 × 40 MIN

MATEMATIKA 6, O'YLAB KO'R

S7 16 INTERPRETING RESULTS

LESSON CLOCK

10'

12'

12'

6'

Four charts, four questions

10 min

Choosing, with a reason

12 min

Charts that mislead

12 min

The task

6 min

LEARNING OBJECTIVES

- Match each kind of data to the chart that shows it best.
- Justify the choice in one sentence.
- Recognise a truncated axis and other misleading devices.
- State the three things every chart must carry.

TEXTBOOK

National

pp. 480–482

Cambridge

Stage 7, pp. 184–188

Workbook

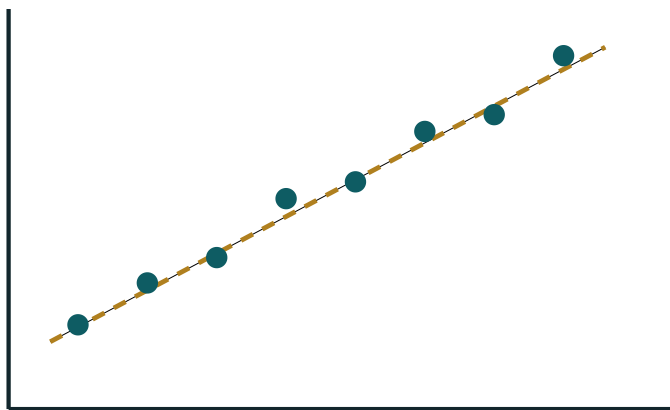
Investigation 16

01

Explanation

Four charts, four questions

CHART	ANSWERS THE QUESTION	GOOD EXAMPLE
bar chart	how do the categories compare in size?	pupils in each of six grades
pie chart	how is one whole shared out?	the school budget by item
line graph	how has one quantity changed over time?	the noon temperature each month
scatter graph	are two measurements related?	height against arm span



positive correlation

A scatter graph — the only one of the four that shows a relationship between two measurements

ASK WHAT THE READER SHOULD SEE AT A GLANCE

A bar chart makes sizes comparable; a pie chart makes shares obvious; a line graph makes a trend visible; a scatter graph makes a relationship visible. Choose the one whose strength is the question.

Choosing, with a reason

THE DATA	BEST CHART	WHY
favourite sport of 60 pupils	pie chart	the 60 form one whole to be shared
pupils in each of six grades	bar chart	six sizes to be compared, not shares
rainfall each month for a year	line graph	the months are in order and the trend matters
height against arm span	scatter graph	two measurements from each pupil
marks out of ten for one class	bar chart of the frequencies	the marks are ordered categories

A LINE GRAPH NEEDS AN ORDERED HORIZONTAL AXIS

Joining “football” to “tennis” with a line says that something exists between them. Nothing does. Time, length and temperature may be joined; favourite sports may not.

Charts that mislead

DEVICE	WHAT THE READER THINKS	THE TRUTH
the vertical axis starts at 40, not 0	one bar is three times another	44 against 52 — a small difference
bars of different widths	the wider bar means more	only the height carries the number
a pie chart with 15 sectors	nothing at all	the thin sectors cannot be read
two pie charts, no totals given	the groups are equal	shares are equal; the numbers may not be
a pictogram with pictures of different sizes	a big picture means many	each picture should stand for a fixed number

THREE THINGS EVERY CHART MUST CARRY

A title that says what is shown; labelled axes or a key; and the total number the data came from. A chart missing any of the three can be read in more than one way.

The task

TODAY'S TASK

For each of these five sets of data, name the chart you would draw and give one sentence of reason — then say what would be lost if the wrong one were chosen instead.

DATA	WRONG CHOICE TO ARGUE AGAINST
the 60 pupils' favourite sports	a line graph
the noon temperature each month	a pie chart
height and arm span of 30 pupils	a bar chart
the population of six grades	a pie chart with no total
a class's marks out of ten	a pie chart with 11 sectors

02 Worked examples**EXAMPLE 1** Sixty pupils name a favourite sport. Which chart, and why?

- ① The 60 pupils make one whole that is being shared out.
- ② A pie chart shows shares directly.
- ③ A bar chart would also work if the question is “which is largest”.
A line graph would not: sports have no order.

ANSWER

A pie chart — the data is shares of one whole

EXAMPLE 2 Monthly rainfall for a year. Which chart, and why?

- ① The months are in order and the interest is the trend.
- ② A line graph joins them meaningfully.
- ③ A pie chart would hide the order entirely.
Which month was wettest would be almost unreadable.

ANSWER

A line graph — ordered time on the horizontal axis

EXAMPLE 3 Two bars stand for 44 and 52, drawn on an axis starting at 40. Why does this mislead?

- ① The drawn heights are 4 and 12 — a ratio of 1 to 3.
- ② The real values differ by under a fifth.
- ③ Starting the axis at 0 shows the truth.
Or mark the break in the axis clearly.

ANSWER

The heights exaggerate the difference by a factor of about three

03 Interactive model

Draw the same two bars twice on the board — once from zero and once from forty. The class sees the trick before it is named.

Which chart, and why

Shares of one whole are best shown by:

a bar chart

a pie chart

a line graph

a scatter graph

Change over time is best shown by:

a pie chart

a line graph

a scatter graph

a pictogram

A relationship between two measurements needs:

a bar chart

a pie chart

a scatter graph

a line graph

Joining “football” to “tennis” with a line is wrong because:

the colours differ

the categories have no order

the totals differ

lines are hard to draw

An axis starting at 40 rather than 0:

saves space honestly

exaggerates differences

is always wrong

changes the data

A pie chart with 15 sectors is:

clearer

unreadable

more accurate

preferred

Bars of different widths suggest:

nothing

that width carries meaning

a larger total

better data

Every chart must carry:

a title, labels and the total

colour

a percentage

a protractor

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Bar chart	Ustunli diagramma	Столбчатая диаграмма
Pie chart	Doiraviy diagramma	Круговая диаграмма
Line graph	Chiziqli grafik	Линейный график
Scatter graph	Sochilish diagrammasi	Диаграмма рассеяния
Category	Toifa	Категория
Axis	O'q	Ось
Misleading	Chalg'ituvchi	Вводящий в заблуждение
Title and key	Sarlavha va izoh	Заголовок и легенда

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A pie chart is best for:

a trend

shares of a whole

two measurements

a timetable

A bar chart compares:

shares

sizes across categories

time only

nothing

A scatter graph needs:

one value each

two values from each person

a total

angles

A truncated vertical axis:

is always fine

exaggerates differences

shrinks differences

has no effect

Two pie charts without totals compare:

numbers

shares only

nothing

means

A pictogram's pictures must:

vary in size

each stand for a fixed number

be coloured

be circles

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The chart for shares of one whole

ANSWER A pie chart

2. The chart for comparing counts across categories

ANSWER A bar chart

3. The chart for change over time

ANSWER A line graph

4. The chart for a relationship between two measurements

ANSWER A scatter graph

5. The sectors of a pie chart total

ANSWER 360°

6. Bars in a bar chart should have

ANSWER Equal widths, with gaps

7. Every chart needs a title, labels and

ANSWER The total the data came from

Medium · the standard question

7 PROBLEMS

1. Favourite sports of 60 pupils

ANSWER A pie chart (or a bar chart)

2. The noon temperature for twelve months

ANSWER A line graph

3. Height against arm span for 30 pupils

ANSWER A scatter graph

4. The number of pupils in each of six grades

ANSWER A bar chart

5. The share of the school budget by item

ANSWER A pie chart

6. A bar chart whose axis starts at 40

ANSWER Misleading

7. Two pie charts of different totals compare

ANSWER Shares, not numbers

Hard · stretch and reason

7 PROBLEMS

1. Why is a pie chart poor for 15 categories?

ANSWER The sectors are too thin to read or label

2. Why is a line graph wrong for favourite sports?

ANSWER The categories have no order

3. Bars of different widths suggest

ANSWER That the area carries meaning, which it does not

4. The best chart for one class's marks out of ten

ANSWER A bar chart of the frequencies

5. The best chart for "do taller pupils have longer arms?"

ANSWER A scatter graph

6. A pie chart of 7 people needs angles to

ANSWER 1 decimal place, adjusted to total 360°

7. Bars of 44 and 52 on an axis from 40

ANSWER Look like 1 to 3

07 Homework

Homework — the task

One page: five choices, five reasons, and one sentence on what the wrong chart would hide.

1. Name the chart you would draw for each of the five data sets in the last section.
2. Give one sentence of reason for each choice.
3. For each, say what would be lost if the wrong chart named beside it were used.
4. Draw the two-bar example twice — from zero and from forty — and label which is honest.
5. List the three things every chart must carry.

Plotting points and drawing graphs of relationships

Coordinates in all four quadrants, and a rule turned into a straight line.
[Cambridge insert]

LESSON 169–170

2 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 11 GRAPHS

LESSON CLOCK

20'

20'

25'

15'

The coordinate grid	20 min
From a rule to a table	20 min
From a table to a line	25 min
Conversion graphs	15 min

LEARNING OBJECTIVES

- Plot and read points in all four quadrants.
- Build a table of values from a rule such as $y = 2x + 1$.
- Plot the table and join the points into a straight line.
- Read values off a graph, including a conversion graph.

TEXTBOOK

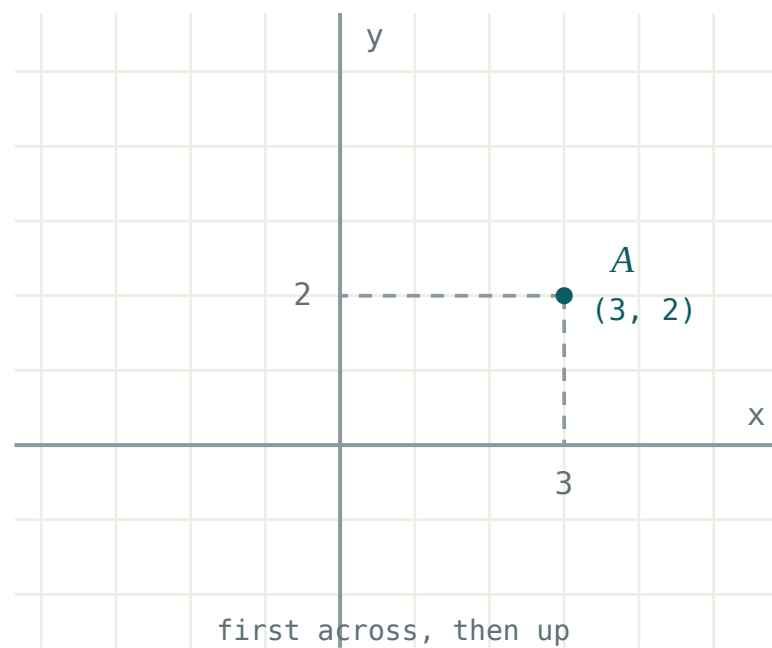
National	pp. 483–487
Cambridge	Stage 7, pp. 112–120
Workbook	Exercise 11.1–11.2

01

Explanation

The coordinate grid

Two number lines crossing at the **origin** fix every point by a pair of numbers: across first, then up. The signs decide the quadrant.



A point fixed by its two coordinates — across first, then up

POINT	SIGNS	QUADRANT	WHERE
(3, 2)	+ , +	first	right and up
(-3, 2)	- , +	second	left and up
(-3, -2)	- , -	third	left and down
(3, -2)	+ , -	fourth	right and down

(3, 2) AND (2, 3) ARE DIFFERENT POINTS

The order is fixed: the first number is the x -coordinate, measured along, and the second is the y -coordinate, measured up. “Along the corridor, then up the stairs.”

From a rule to a table

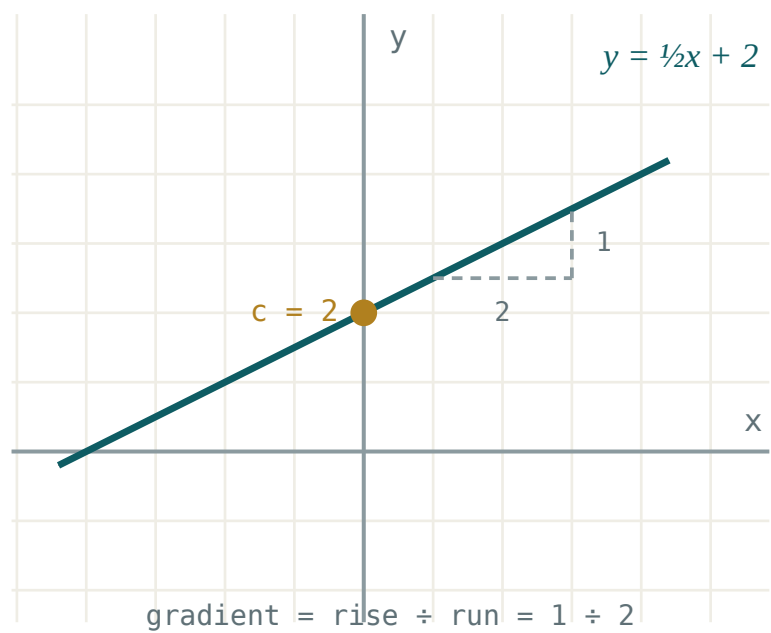
A rule connects two quantities. Choose values of x , work out each y , and collect the pairs in a table.

X	-2	-1	0	1	2	3
$y = 2x + 1$	-3	-1	1	3	5	7
$y = x + 3$	1	2	3	4	5	6
$y = 10 - x$	12	11	10	9	8	7

EQUAL STEPS IN x GIVE EQUAL STEPS IN y

In the first row every step of 1 across raises y by 2 ; in the last it lowers y by 1 . That steadiness is exactly why the points will lie on a straight line.

From a table to a line



A straight-line graph — the steady climb and the point where it crosses the y -axis

STEP	WHAT TO DO	WHY
1	draw and number both axes, including the negatives	every point must fit
2	plot each pair from the table with a small cross	a dot hides its own centre
3	check the points look like a line before joining	one stray point is an arithmetic slip
4	join them with a ruler and extend a little each way	the rule holds beyond the table
5	label the line with its rule	two lines on one grid must be told apart

A POINT OUT OF LINE IS A MISTAKE, NOT A FEATURE

Recalculate that y rather than drawing a bend. The rules in this topic all give perfectly straight lines.

Conversion graphs

A straight-line graph is the fastest way to convert between two units. Eight kilometres is about five miles, so the line through $(0, 0)$ and $(8, 5)$ converts either way.

KILOMETRES	8	16	24	40
miles	5	10	15	25

Not every real-life graph starts at the origin. A taxi charging 5000 so'm plus 2000 so'm a kilometre has the rule $c = 5000 + 2000d$.

KILOMETRES d	0	1	3	5	7
cost in so'm	5000	7000	11 000	15 000	19 000

READ ACROSS, THEN DOWN – OR DOWN, THEN ACROSS

To convert 24 km, go up from 24 to the line and across to 15 miles. To go the other way, start on the miles axis. One line answers both questions.

02 Worked examples

EXAMPLE 1 Complete a table for $y = 2x + 1$ from $x = -2$ to $x = 3$.

- ① Double each x and add 1.

- ② $-3, -1, 1, 3, 5, 7$.

- ③ Each step of 1 in x raises y by 2.
So the points lie on a line ✓

ANSWER

$-3, -1, 1, 3, 5, 7$

EXAMPLE 2 Using $8 \text{ km} = 5 \text{ miles}$, convert 24 km and 20 miles .

- ① 24 km is three lots of 8.

- ② So $3 \cdot 5 = 15 \text{ miles}$.

- ③ 20 miles is four lots of 5, so $4 \cdot 8 = 32 \text{ km}$.
The same line, read both ways.

ANSWER

15 miles and 32 km

EXAMPLE 3 A taxi charges 5000 so‘m plus 2000 so‘m a kilometre. A ride costs 19 000 so‘m. How far was it?

① $19\,000 - 5000 = 14\,000$ so‘m for the distance.

② $14\,000 \div 2000$.

③ $= 7$ km.

Check: $5000 + 14\,000 = 19\,000$ ✓

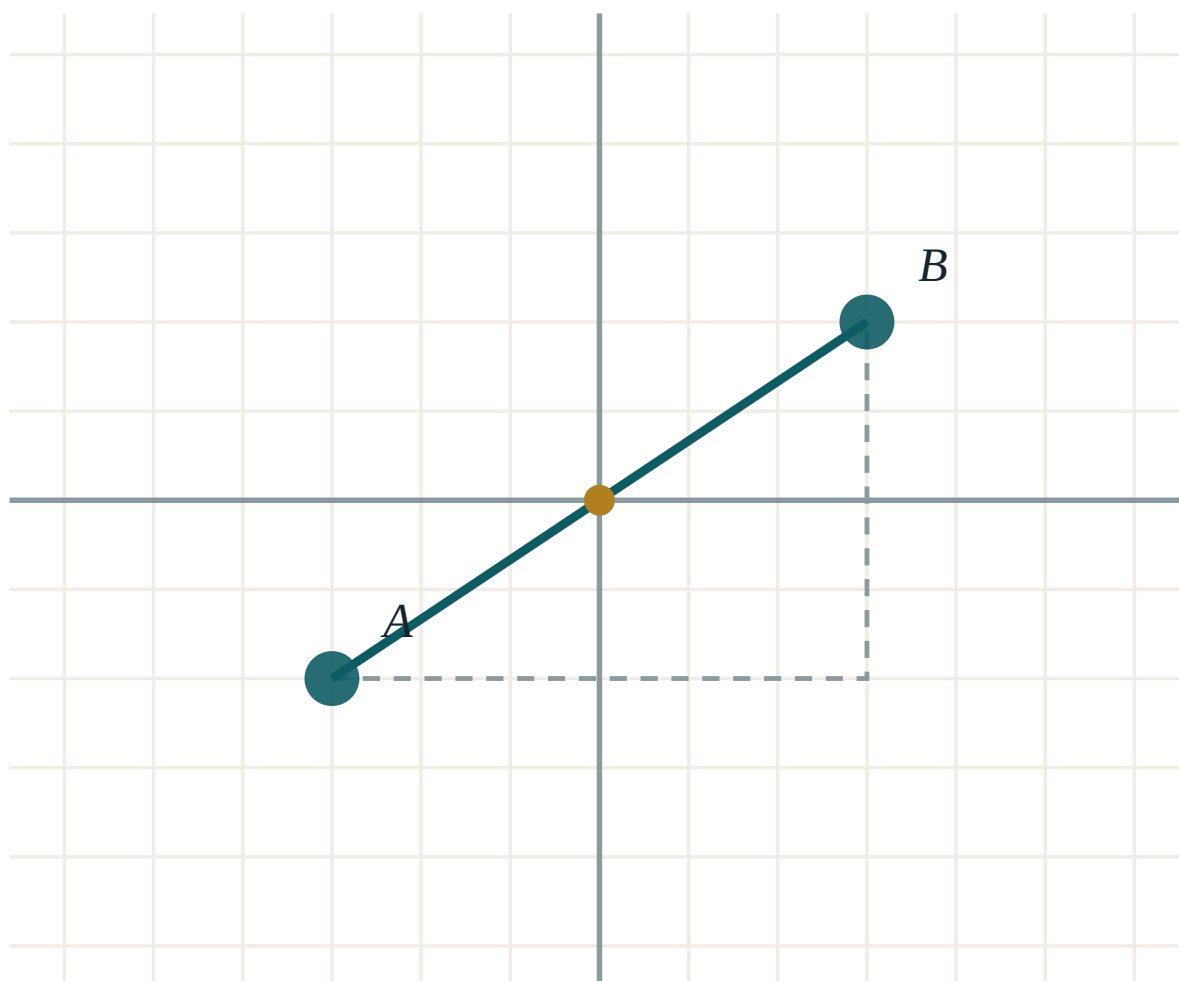
ANSWER

7 km

03 Interactive model

Drag one point in the model along a row and watch only the first coordinate change; that is the whole of “across first, then up”. The distance readout is a Grade 8 preview — ignore it for now.

Points on the coordinate grid



A (-3, -2)

B (3, 2)

 $x_2 - x_1$ 6 $y_2 - y_1$ 4

distance AB 7.21

midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Coordinates	Koordinatalar	Координаты
Origin	Koordinata boshi	Начало координат
x-axis	Abssissa o'qi	Ось абсцисс
y-axis	Ordinata o'qi	Ось ординат
Quadrant	Chorak	Четверть
Table of values	Qiyimatlar jadvali	Таблица значений
Straight-line graph	To'g'ri chiziqli grafik	Прямолинейный график
Conversion graph	O'tkazish grafigi	График перевода

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

In $(3, -2)$ the -2 is measured:

along

up or down

diagonally

from B

$(-3, 2)$ lies in the:

first quadrant

second quadrant

third quadrant

fourth quadrant

$y = 2x + 1$ at $x = 3$ is:

5

7

6

9

A graph crosses the y -axis where:

Points from $y = 2x + 1$ lie on:

With $8 \text{ km} = 5 \text{ miles}$, 40 km is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The quadrant of $(3, 2)$

ANSWER The first

2. The quadrant of $(-3, 2)$

ANSWER The second

3. The quadrant of $(-3, -2)$

ANSWER The third

4. The quadrant of $(3, -2)$

ANSWER The fourth

5. $y = 2x + 1$ at $x = 3$

ANSWER 7

6. $y = 2x + 1$ at $x = 0$

ANSWER 1

7. On the y -axis the x -coordinate is

ANSWER 0

Medium · the standard question

7 PROBLEMS

1. $y = 2x + 1$ for $x = -2, -1, 0, 1$

ANSWER $-3, -1, 1, 3$

2. $y = x + 3$ at $x = -2$

ANSWER 1

3. $y = 3x - 2$ at $x = 2$

ANSWER 4

4. $y = 10 - x$ at $x = 4$

ANSWER 6

5. With $8 \text{ km} = 5 \text{ miles}$, 16 km

ANSWER 10 miles

6. ...and 40 km

ANSWER 25 miles

7. A taxi at $5000 + 2000d$ for 3 km

ANSWER $11\,000 \text{ so'm}$

Hard · stretch and reason

7 PROBLEMS

1. $y = 2x + 1$: the value of x when $y = 11$

ANSWER $x = 5$

2. $y = 3x - 2$: the value of x when $y = 13$

ANSWER $x = 5$

3. With $8 \text{ km} = 5 \text{ miles}$, 24 km

ANSWER 15 miles

4. ...and 20 miles

ANSWER 32 km

5. A taxi ride at $5000 + 2000d$ costs $19\,000$ so'm: the distance

ANSWER 7 km

6. Why do the points of $y = 2x + 1$ lie on a line?

ANSWER Equal steps in x give equal steps in y

7. The graph of $y = 10 - x$ slopes

ANSWER Downwards

07 Homework

Homework — 5 tasks

Plot with small crosses, join with a ruler, and label each line with its rule.

- Plot $(2, 3)$, $(-2, 3)$, $(-2, -3)$ and $(2, -3)$, and name the shape they make.
- Complete a table for $y = 3x - 1$ from $x = -2$ to $x = 3$, and draw the graph.
- On the same grid draw $y = x + 3$ and write down where the two lines cross.
- Using $8 \text{ km} = 5 \text{ miles}$, convert 32 km and 15 miles .

5. A taxi charges 4000 so'm plus 1500 so'm a kilometre. Find the cost of 6 km and the distance costing 16 000 so'm.

The probability scale; calculating and estimating probability

From “likely” to a number between 0 and 1 — by counting, and by experiment.
[Cambridge insert]

LESSON 171–172

2 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 13 PROBABILITY

LESSON CLOCK

20'

22'

20'

18'

The probability scale	20 min
Counting the outcomes	22 min
Probabilities that add to 1	20 min
Estimating from experiment	18 min

LEARNING OBJECTIVES

- Place an event on the probability scale from 0 to 1.
- Calculate a probability when the outcomes are equally likely.
- Use the fact that all the probabilities of one experiment add to 1.
- Estimate a probability from the results of an experiment.

TEXTBOOK

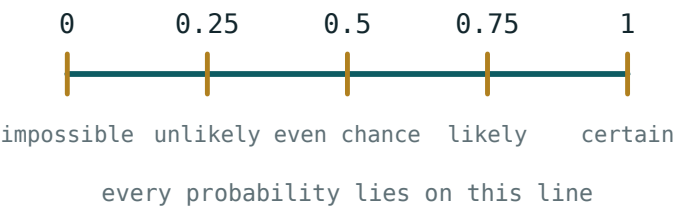
National	pp. 488–492
Cambridge	Stage 7, pp. 132–140
Workbook	Exercise 13.1–13.2

01

Explanation

The probability scale

Every probability is a number from 0 to 1. Nothing is less likely than impossible and nothing more likely than certain, so nothing lies outside.



The probability scale — impossible at 0, certain at 1

EVENT	IN WORDS	PROBABILITY
a die shows 7	impossible	0
a coin shows a head	even chance	$\frac{1}{2}$
a die shows a number under 7	certain	1
a die shows 4	unlikely	$\frac{1}{6}$

A PROBABILITY IS NEVER A NUMBER LIKE 2 OR -0.3

An answer outside 0 to 1 means the counting went wrong — usually the wrong total underneath. Check that number before anything else.

Counting the outcomes

When every outcome is equally likely, a probability is a fraction: how many outcomes count as success, over how many there are altogether.

$$P(event) = \frac{\text{favourable outcomes}}{\text{all outcomes}}$$

EXPERIMENT	EVENT	COUNT	PROBABILITY
one fair die	a 4	1 of 6	$\frac{1}{6}$
one fair die	an even number	3 of 6	$\frac{1}{2}$
one fair die	more than 4	2 of 6	$\frac{1}{3}$
one fair die	a multiple of 3	2 of 6	$\frac{1}{3}$
a bag of 10 counters	red, with 4 red	4 of 10	$\frac{2}{5}$

“EQUALLY LIKELY” IS A REAL CONDITION, NOT A FORMALITY

The fraction rule works for a fair die and for counters drawn without looking. It does not work for “it rains tomorrow”, where the two outcomes are not equally likely at all.

Probabilities that add to 1

A bag holds 4 red, 3 blue, 2 green and 1 yellow counter.

COLOUR	COUNTERS	PROBABILITY	IN LOWEST TERMS
red	4	$\frac{4}{10}$	$\frac{2}{5}$
blue	3	$\frac{3}{10}$	$\frac{3}{10}$
green	2	$\frac{2}{10}$	$\frac{1}{5}$
yellow	1	$\frac{1}{10}$	$\frac{1}{10}$
total	10	$\frac{10}{10}$	1

$$P(\text{not red}) = 1 - P(\text{red}) = 1 - \frac{2}{5} = \frac{3}{5}$$

SUBTRACTING FROM ONE SAVES A LOT OF COUNTING

“Not a six” has five favourable outcomes, but $1 - \frac{1}{6} = \frac{5}{6}$ is quicker and safer — and for a large set of outcomes it is the only sensible route.

Estimating from experiment

Some probabilities cannot be counted, because the outcomes are not equally likely. Then the experiment itself is the evidence: the **relative frequency** estimates the probability.

$$\text{relative frequency} = \frac{\text{times it happened}}{\text{times it was tried}}$$

EXPERIMENT	RESULT	ESTIMATE	COMMENT
a drawing pin dropped 100 times	point up 63	0.63	no counting could have found it
a die thrown 60 times	12 sixes	0.2	theory says $\frac{1}{6} \approx 0.167$
a coin tossed 200 times	92 heads	0.46	close to 0.5 — it looks fair

MORE TRIALS, BETTER ESTIMATE

Ten throws of a fair die can easily give four sixes; six hundred throws will not. A relative frequency far from the theory is evidence of bias only when the number of trials is large.

Once a probability is known, the expected number in n trials is n times it: 300 throws of a fair die should give about $300 \cdot \frac{1}{6} = 50$ sixes.

02

Worked examples

EXAMPLE 1 A bag holds 4 red, 3 blue, 2 green and 1 yellow counter. Find $P(\text{red})$ and $P(\text{not red})$.

1

There are 10 counters altogether.

2

$P(\text{red}) = \frac{4}{10} = \frac{2}{5}$.

3

$P(\text{not red}) = 1 - \frac{2}{5} = \frac{3}{5}$.

Check: 6 of the 10 are not red ✓

ANSWER

$\frac{2}{5}$ and $\frac{3}{5}$

EXAMPLE 2 One fair die is thrown. Find $P(\text{even})$, $P(\text{more than 4})$ and $P(7)$.

① Even: 2, 4, 6 — three of six, so $\frac{1}{2}$.

② More than 4: 5, 6 — two of six, so $\frac{1}{3}$.

③ A 7 is impossible, so 0.
All three lie between 0 and 1 ✓

ANSWER

$\frac{1}{2}$, $\frac{1}{3}$ and 0

EXAMPLE 3 A drawing pin lands point up 63 times in 100 drops. Estimate the probability and the number of point-up landings in 500 drops.

① Relative frequency $\frac{63}{100} = 0.63$.

② Expected number = $500 \cdot 0.63$.

③ = 315.
An estimate, not a guarantee.

ANSWER

0.63 and about 315

03 Interactive model

Throw a die twenty times as a class and write the running relative frequency of sixes on the board; watching it settle towards a sixth is worth more than any explanation.

Numbers between 0 and 1

An impossible event has probability:

$P(4)$ on a fair die is:

$P(\text{even})$ on a fair die is:

$P(\text{more than } 4)$ on a fair die is:

$$\frac{1}{6}$$

$$\frac{1}{3}$$

$$\frac{1}{2}$$

$$\frac{2}{3}$$

With 4 red in 10 counters, $P(\text{not red})$ is:

$$\frac{2}{5}$$

$$\frac{3}{5}$$

$$\frac{1}{2}$$

$$\frac{6}{4}$$

All the probabilities of one experiment add to:

$$0$$

$$\frac{1}{2}$$

$$1$$

$$100$$

12 sixes in 60 throws gives a relative frequency of:

0.12

0.2

0.6

5

In 300 throws of a fair die the expected number of sixes is:

6

30

50

60

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Probability	Ehtimollik	Вероятность
Event	Hodisa	Событие
Outcome	Natija	Исход
Equally likely	Teng ehtimolli	Равновозможные
Impossible	Imkonsiz	Невозможное
Certain	Muqarrar	Достоверное
Relative frequency	Nisbiy chastota	Относительная частота
Fair	Adolatli (bir jinsli)	Правильный (симметричный)

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Every probability lies:

above 1

between 0 and 1

below 0

anywhere

A certain event has probability:

0

$\frac{1}{2}$

1

100

The fraction rule needs the outcomes to be:

many

equally likely

few

written down

$P(\text{not } A)$ equals:

$$1 - P(A)$$

$$P(A) - 1$$

$$1 \div P(A)$$

$$0$$

Relative frequency is:

a theory

an estimate from experiment

always exact

a percentage

A better estimate comes from:

fewer trials

more trials

a bigger die

rounding

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The probability of an impossible event

ANSWER 0

2. The probability of a certain event

ANSWER 1

3. $P(\text{head})$ on a fair coin

ANSWER $\frac{1}{2}$

4. $P(4)$ on a fair die

ANSWER $\frac{1}{6}$

5. $P(\text{even})$ on a fair die

ANSWER $\frac{1}{2}$

6. $P(7)$ on a fair die

ANSWER 0

7. Every probability lies between

ANSWER 0 and 1

Medium · the standard question

7 PROBLEMS

1. $P(\text{more than } 4)$ on a fair die

ANSWER $\frac{1}{3}$

2. $P(\text{red})$ with 4 red in 10

ANSWER $\frac{2}{5}$

3. $P(\text{blue})$ with 3 blue in 10

ANSWER $\frac{3}{10}$

4. $P(\text{not red})$ for that bag

ANSWER $\frac{3}{5}$

5. The four probabilities of that bag add to

ANSWER 1

6. A pin lands point up 63 times in 100

ANSWER 0.63

7. 12 sixes in 60 throws

ANSWER 0.2

Hard · stretch and reason

7 PROBLEMS

1. $P(\text{a multiple of } 3)$ on a fair die

ANSWER $\frac{1}{3}$

2. $P(\text{not a } 6)$ on a fair die

ANSWER $\frac{5}{6}$

3. 5 red and 15 blue: $P(\text{red})$ as a percentage

ANSWER 25%

4. Is a die giving 12 sixes in 60 throws unfair?

ANSWER Probably not — 10 expected, 12 seen

5. Expected sixes in 300 throws of a fair die

ANSWER 50

6. $P(\text{red}) = 0.4$ in a bag of 25: the number of red

ANSWER 10

7. If $P(\text{win}) = 0.35$, then $P(\text{not win})$

ANSWER 0.65

07 Homework

Homework — 5 tasks

Every answer as a fraction in lowest terms or a decimal, and never outside 0 to 1.

1. A fair die is thrown. Find $P(3)$, $P(\text{odd})$ and $P(\text{at least } 5)$.
2. A bag holds 6 red, 9 blue and 5 green counters. Find $P(\text{red})$, $P(\text{blue})$ and $P(\text{not green})$.
3. Show that the three probabilities in task 2 add to 1.
4. A coin lands heads 92 times in 200 tosses. Estimate $P(\text{head})$ and say whether the coin looks fair.

5. If $P(\text{rain}) = 0.3$, find $P(\text{no rain})$ and the expected number of rainy days in 30.

Reflection, rotation and translation on a grid

Three ways of moving a shape that leave every length and every angle alone.
[Cambridge insert]

LESSON 173–174

2 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 14 POSITION AND TRANSFORMATION

LESSON CLOCK

16'

22'

24'

18'

The three transformations	16 min
Reflection	22 min
Rotation	24 min
Translation	18 min

LEARNING OBJECTIVES

- Reflect a shape in the axes and in the line $y = x$.
- Rotate a shape about the origin through 90° and 180° .
- Translate a shape by a given pair of numbers.
- State what each transformation keeps and what it changes.

TEXTBOOK

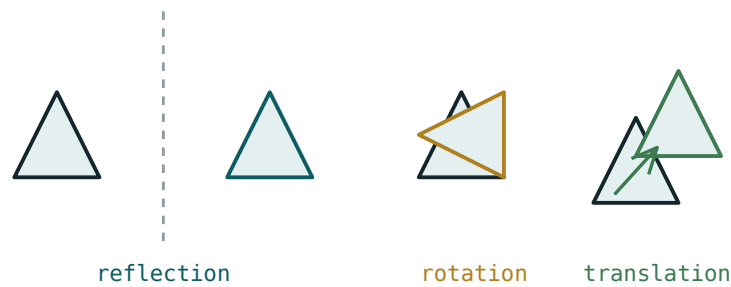
National	pp. 493–497
Cambridge	Stage 7, pp. 144–152
Workbook	Exercise 14.1–14.3

01

Explanation

The three transformations

Each of the three moves a shape without bending or resizing it: every length and every angle of the **image** matches the original. Only the position — and for a reflection the sense — is different.



Reflection, rotation and translation of the same triangle

TRANSFORMATION	NEEDS	LENGTHS	ORIENTATION
reflection	a mirror line	unchanged	reversed
rotation	a centre, an angle and a direction	unchanged	kept
translation	two numbers — across and up	unchanged	kept

DESCRIBING A TRANSFORMATION FULLY IS PART OF THE ANSWER

“A rotation” is incomplete; “a rotation of 90° anticlockwise about the origin” is not. A reflection needs its mirror line, a translation its two numbers.

Reflection

Each point moves straight across the mirror line to the same distance on the other side. A point already on the line does not move at all.

MIRROR	RULE	$A(1, 1)$	$B(4, 1)$	$C(2, 3)$
the y-axis	$(x, y) \rightarrow (-x, y)$	$(-1, 1)$	$(-4, 1)$	$(-2, 3)$
the x-axis	$(x, y) \rightarrow (x, -y)$	$(1, -1)$	$(4, -1)$	$(2, -3)$
the line $y = x$	$(x, y) \rightarrow (y, x)$	$(1, 1)$	$(1, 4)$	$(3, 2)$

A REFLECTION TURNS THE SHAPE OVER

Label the vertices A, B, C and read them round the triangle: if they ran clockwise before, they run anticlockwise after. Nothing else in this topic does that.

Rotation

Every point swings round the centre through the same angle, staying the same distance from it. About the origin the three useful cases have simple rules.

ROTATION ABOUT O	RULE	$A(1, 1)$	$B(4, 1)$	$C(2, 3)$
90°	anticlockwise: $(x, y) \rightarrow (-y, x)$	$(-1, 1)$	$(-1, 4)$	$(-3, 2)$
180°	either way: $(x, y) \rightarrow (-x, -y)$	$(-1, -1)$	$(-4, -1)$	$(-2, -3)$
90°	clockwise: $(x, y) \rightarrow (y, -x)$	$(1, -1)$	$(1, -4)$	$(3, -2)$

THE CENTRE IS THE ONLY POINT THAT STAYS STILL

For a rotation of 180° the direction does not matter; for 90° it decides everything, so it must be stated. Tracing paper turned about a pin is the honest way to check.

Translation

Every point slides the same distance in the same direction — so many across, so many up. Nothing turns and nothing flips.

TRANSLATION	RULE	$A(1, 1)$	$B(4, 1)$	$C(2, 3)$
3 right, 2 down	$(x, y) \rightarrow (x + 3, y - 2)$	$(4, -1)$	$(7, -1)$	$(5, 1)$
2 left, 3 up	$(x, y) \rightarrow (x - 2, y + 3)$	$(-1, 4)$	$(2, 4)$	$(0, 6)$

EVERY TRANSFORMATION HERE CAN BE UNDONE

A translation of 3 right and 2 down is undone by 3 left and 2 up; a reflection is undone by itself; a rotation of 90° anticlockwise by 90° clockwise. Two reflections, in the x -axis then the y -axis, make a rotation of 180° .

02 Worked examples

EXAMPLE 1 Reflect the triangle $A(1, 1)$, $B(4, 1)$, $C(2, 3)$ in the y -axis.

- 1 The rule is $(x, y) \rightarrow (-x, y)$.
Only the sign of x changes.
- 2 $A(1, 1) \rightarrow (-1, 1)$, $B(4, 1) \rightarrow (-4, 1)$.
- 3 $C(2, 3) \rightarrow (-2, 3)$.
The labels now read the other way round ✓

ANSWER

$(-1, 1)$, $(-4, 1)$ and $(-2, 3)$

EXAMPLE 2 Rotate the same triangle 90° anticlockwise about the origin.

- 1 The rule is $(x, y) \rightarrow (-y, x)$.
- 2 $A(1, 1) \rightarrow (-1, 1)$, $B(4, 1) \rightarrow (-1, 4)$.
- 3 $C(2, 3) \rightarrow (-3, 2)$.
Distances from O are unchanged ✓

ANSWER

$(-1, 1)$, $(-1, 4)$ and $(-3, 2)$

EXAMPLE 3 Translate the same triangle 3 right and 2 down.

① Add 3 to each x and subtract 2 from each y .

② $(4, -1), (7, -1)$.

③ $(5, 1)$.

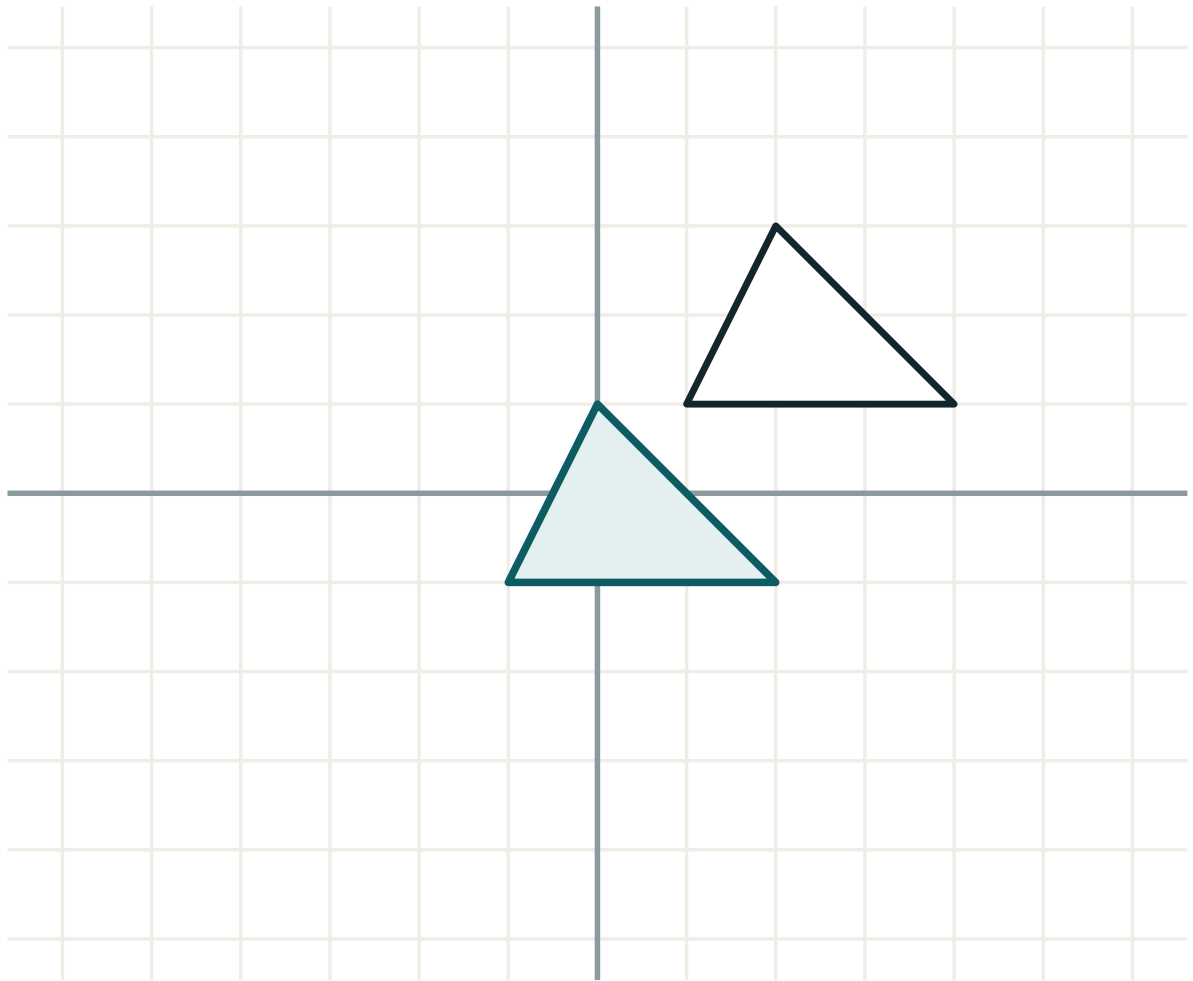
The shape has not turned at all ✓

ANSWER

$(4, -1), (7, -1)$ and $(5, 1)$

03 Interactive model

Switch the model between reflection and rotation while watching the orientation readout; that one word is the difference the class most often misses.

Move the shapetransformation **translation**lengths **unchanged**angles **unchanged**orientation **kept****04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Transformation	Almashtirish	Преобразование
Reflection	Simmetriya (aks ettirish)	Отражение
Mirror line	Simmetriya o'qi	Ось симметрии
Rotation	Burish	Поворот
Centre of rotation	Burilish markazi	Центр поворота
Translation	Parallel ko'chirish	Параллельный перенос
Image	Tasvir	Образ
Orientation	Yo'nalish	Ориентация

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Reflection in the y -axis sends $(3, 2)$ to:

A rotation of 180° about O sends $(3, 2)$ to:

Which transformation reverses the orientation?

A rotation must be described with:

a mirror line

a centre, an angle and a direction

two numbers

nothing

A translation of 3 right and 2 down is undone by:

3 right, 2 up

3 left, 2 up

2 left, 3 up

itself

Under all three transformations the lengths are:

doubled

halved

unchanged

reversed

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Reflection in the y -axis sends $(3, 2)$ to

ANSWER $(-3, 2)$

2. Reflection in the x -axis sends $(3, 2)$ to

ANSWER $(3, -2)$

3. A translation of 2 right and 3 up sends $(1, 1)$ to

ANSWER $(3, 4)$

4. A rotation of 180° about O sends $(3, 2)$ to

ANSWER $(-3, -2)$

5. All three transformations keep

ANSWER The lengths and the angles

6. A reflection needs

ANSWER A mirror line

7. A rotation needs

ANSWER A centre, an angle and a direction

Medium · the standard question

7 PROBLEMS

1. Reflection in $y = x$ sends $(4, 1)$ to

ANSWER $(1, 4)$

2. A 90° anticlockwise rotation about O sends $(4, 1)$ to

ANSWER $(-1, 4)$

3. A 90° clockwise rotation about O sends $(4, 1)$ to

ANSWER $(1, -4)$

4. A translation of 3 right and 2 down sends $(2, 3)$ to

ANSWER $(5, 1)$

5. Reflect $A(1, 1)$, $B(4, 1)$, $C(2, 3)$ in the y -axis

ANSWER $(-1, 1)$, $(-4, 1)$, $(-2, 3)$

6. Which transformation reverses the labels round the shape?

ANSWER Reflection

7. A translation is described by

ANSWER Two numbers — across and up

Hard · stretch and reason

7 PROBLEMS

1. Rotate $A(1, 1)$, $B(4, 1)$, $C(2, 3)$ by 180° about O

ANSWER $(-1, -1)$, $(-4, -1)$, $(-2, -3)$

2. Reflection in the x -axis then the y -axis equals

ANSWER A rotation of 180° about O

3. The translation that undoes 3 right and 2 down

ANSWER 3 left and 2 up

4. The rotation that undoes 90° anticlockwise

ANSWER 90° clockwise

5. A point on the mirror line maps to

ANSWER Itself

6. Two rotations of 90° anticlockwise about O make

ANSWER A rotation of 180°

7. Which transformation leaves no point fixed?

ANSWER A translation

07 Homework

Homework — 5 tasks

Describe every transformation fully, and label the image A' , B' , C' .

- Plot $A(1, 2)$, $B(4, 2)$, $C(1, 4)$ and reflect the triangle in the y -axis.
- Reflect the original triangle in the x -axis and write the three image points.
- Rotate the original triangle 90° clockwise about the origin.
- Translate the original triangle 4 left and 3 down.
- State which of your four images have the reversed orientation, and explain why.

Recall — three-dimensional shapes

Naming the solids, counting their faces, edges and vertices, and a pattern that links the three.

LESSON 175

1 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 8.4 3-D SHAPES

LESSON CLOCK

10'

12'

12'

6'

Naming the solids	10 min
Faces, edges and vertices	12 min
The pattern $F + V - E$	12 min
Prisms and pyramids in general	6 min

LEARNING OBJECTIVES

- Name the common solids and say which have curved surfaces.
- Count the faces, edges and vertices of a prism and a pyramid.
- Use the pattern $F + V - E = 2$ as a check.
- Give the counts for a prism or pyramid on an n -sided base.

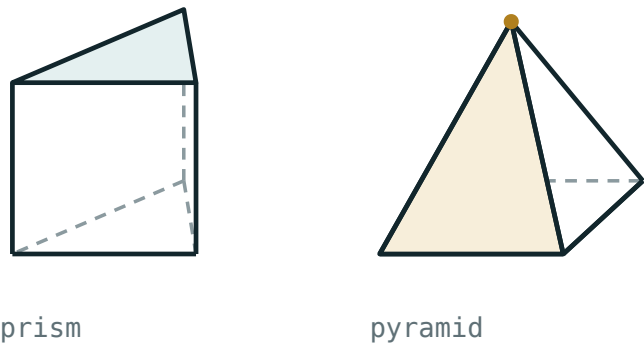
TEXTBOOK

National	pp. 498–501
Cambridge	Stage 7, pp. 84–88
Workbook	Exercise 8.4

01

Explanation

Naming the solids



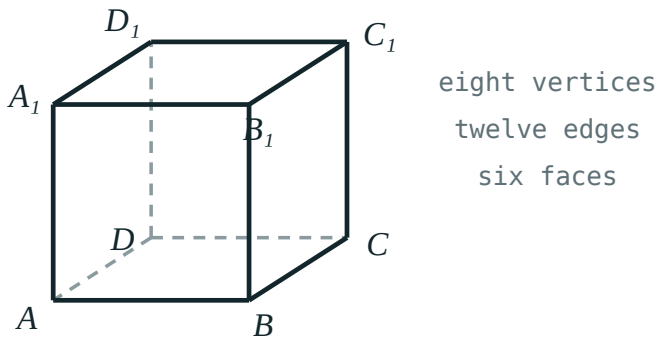
A prism and a pyramid — the two families that most solids belong to

SOLID	FLAT FACES	CURVED SURFACES	RECOGNISE IT BY
cube and cuboid	6	0	six rectangles
prism	$n + 2$	0	two identical ends, joined by rectangles
pyramid	$n + 1$	0	a base and triangles meeting at an apex
cylinder	2	1	two circles and a tube
cone	1	1	a circle and an apex
sphere	0	1	every point the same distance from the centre

A PRISM IS NAMED BY ITS CROSS-SECTION

Cut a prism anywhere across and the face is the same shape every time — triangular, pentagonal, hexagonal. A cuboid is simply a rectangular prism.

Faces, edges and vertices



Six faces, twelve edges, eight vertices

SOLID	FACES F	EDGES E	VERTICES V
cube or cuboid	6	12	8
triangular prism	5	9	6
pentagonal prism	7	15	10
square pyramid	5	8	5
triangular pyramid	4	6	4
hexagonal pyramid	7	12	7

COUNT THE HIDDEN ONES TOO

A drawing shows at most three faces of a cube, but there are six. Turning the solid — or the drawing — in the mind is the skill this recall is really about.

The pattern $F + V - E$

Add the faces and the vertices, subtract the edges, and every solid with flat faces gives the same answer.

$F + V - E = 2$

SOLID	$F + V$	E	$F + V - E$
cube	14	12	2
triangular prism	11	9	2
square pyramid	10	8	2
hexagonal prism	20	18	2
triangular pyramid	8	6	2

USE IT TO CHECK A COUNT, NOT TO MAKE ONE

If your three numbers do not give 2, one of them is wrong — and it is almost always the edges, because some of them are hidden. The rule does not apply to a cylinder, a cone or a sphere: their surfaces are not made of flat polygons.

Prisms and pyramids in general

SOLID ON AN n -SIDED BASE	FACES	EDGES	VERTICES
prism	$n + 2$	$3n$	$2n$
pyramid	$n + 1$	$2n$	$n + 1$

BOTH FORMULAS PASS THE CHECK

For the prism, $(n + 2) + 2n - 3n = 2$. For the pyramid, $(n + 1) + (n + 1) - 2n = 2$. With $n = 6$ they give 8, 18, 12 and 7, 12, 7 — the last two rows of the previous table.

02 Worked examples

EXAMPLE 1 Count the faces, edges and vertices of a triangular prism, and check the pattern.

1

Two triangular ends and three rectangles: $F = 5$.

2

Three edges at each end and three joining them: $E = 9$.

3

Three vertices at each end: $V = 6$; $5 + 6 - 9 = 2$ ✓

ANSWER

5 faces, 9 edges, 6 vertices

EXAMPLE 2 A pyramid has a hexagonal base. Find its faces, edges and vertices.

- ① The base and six triangles: $F = 7$.
- ② Six base edges and six sloping ones: $E = 12$.
- ③ Six base vertices and the apex: $V = 7$; $7 + 7 - 12 = 2$ ✓

ANSWER

7 faces, 12 edges, 7 vertices

EXAMPLE 3 A pupil counts 8 faces, 12 vertices and 16 edges for a hexagonal prism. Find the mistake.

- ① $8 + 12 - 16 = 4$, not 2.
So a count is wrong.
- ② A hexagonal prism has $3n = 18$ edges.
Six top, six bottom, six upright.
- ③ The two missing edges are the hidden uprights.
 $8 + 12 - 18 = 2$ ✓

ANSWER

The edges: there are 18, not 16

03 Interactive model

Pass round a real prism and a real pyramid while the counting is done; the hidden edges stop being hidden the moment the solid can be turned over.

Counting on a solid

A cube has how many edges?

A triangular prism has how many faces?

A square pyramid has how many vertices?

$F + V - E$ for any solid with flat faces is:

0

1

2

it varies

A cylinder has how many curved surfaces?

0

1

2

3

A cone has how many flat faces?

0

1

2

3

A prism on an n -sided base has how many edges?

$2n$

$3n$

$n + 2$

$n + 1$

A pyramid on an n -sided base has how many faces?

n

$n + 1$

$n + 2$

$2n$

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Solid	Jism	Тело
Prism	Prizma	Призма
Pyramid	Piramida	Пирамида
Cylinder	Silindr	Цилиндр
Cone	Konus	Конус
Sphere	Shar	Шар
Apex	Uchi (cho'qqi)	Вершина (апекс)
Curved surface	Egri sirt	Криволинейная поверхность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A prism is named by:

its height

its cross-section

its colour

its volume

A sphere has how many flat faces?

0

1

2

6

A triangular pyramid has how many faces?

3

4

5

6

If your counts give $F + V - E = 4$ then:

the solid is unusual

a count is wrong

the rule fails

nothing

Euler's pattern does not apply to:

a prism

a pyramid

a cylinder

a cube

A cuboid is:

a pyramid

a rectangular prism

a cylinder

a cone

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The faces of a cube

ANSWER 6

2. The edges of a cube

ANSWER 12

3. The vertices of a cube

ANSWER 8

4. The faces of a triangular prism

ANSWER 5

5. The edges of a triangular prism

ANSWER 9

6. The vertices of a triangular prism

ANSWER 6

7. The flat faces of a sphere

ANSWER 0

Medium · the standard question

7 PROBLEMS

1. Faces, edges and vertices of a square pyramid

ANSWER 5, 8, 5

2. Faces, edges and vertices of a triangular pyramid

ANSWER 4, 6, 4

3. Faces, edges and vertices of a pentagonal prism

ANSWER 7, 15, 10

4. $F + V - E$ for a cube

ANSWER 2

5. $F + V - E$ for a square pyramid

ANSWER 2

6. The curved surfaces of a cylinder

ANSWER 1

7. The apexes of a cone

ANSWER 1

Hard · stretch and reason

7 PROBLEMS

1. Faces, edges and vertices of a hexagonal prism

ANSWER 8, 18, 12

2. Faces, edges and vertices of a hexagonal pyramid

ANSWER 7, 12, 7

3. The faces of a prism on an n -sided base

ANSWER $n + 2$

4. Its edges

ANSWER $3n$

5. Its vertices

ANSWER $2n$

6. The edges of a pyramid on an n -sided base

ANSWER $2n$

7. Why does the pattern fail for a cylinder?

ANSWER Its surface is not made of flat polygons

07 Homework

Homework — 5 tasks

Check every set of three counts with $F + V - E = 2$ before writing it down.

1. Count the faces, edges and vertices of a pentagonal pyramid and check the pattern.
2. Do the same for an octagonal prism.
3. Name three solids with a curved surface and say how many flat faces each has.
4. A solid has 9 faces and 9 vertices. How many edges must it have?
5. Explain in one sentence why a cuboid is a kind of prism.

The cone, the cylinder, the prism and the pyramid

Four families of solid, told apart by their cross-sections — and one volume rule that covers two of them.

LESSON 176–178

3 × 40 MIN

MATEMATIKA 6, §34

S7 8.4 3-D SHAPES

LESSON CLOCK

30'

30'

30'

25'

5

Prisms and the cross-section	30 min
The cylinder	30 min
Pyramids and the cone	30 min
The volume of a prism	25 min
Homework	5 min

LEARNING OBJECTIVES

- Describe a prism by its cross-section and count its parts.
- Describe a cylinder, a cone and a pyramid, naming their surfaces.
- Find the volume of a prism as base area times height.
- Find the volume and the curved surface of a cylinder.

TEXTBOOK

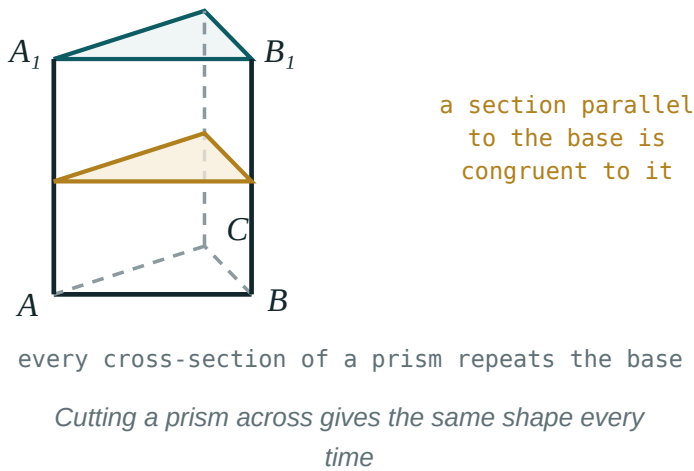
National	pp. 502–511
Cambridge	Stage 7, pp. 84–90
Workbook	Exercise 8.4

01

Explanation

Prisms and the cross-section

A prism has two identical ends joined by rectangles. Slice it anywhere across and the face you expose is always that same shape — its **cross-section**.



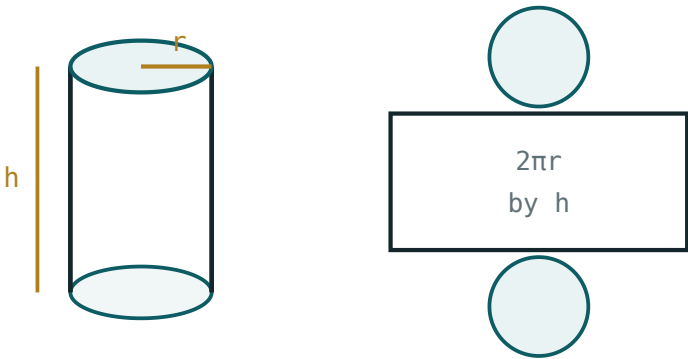
PRISM	CROSS-SECTION	FACES	EDGES	VERTICES
cuboid	rectangle	6	12	8
triangular prism	triangle	5	9	6
pentagonal prism	pentagon	7	15	10
hexagonal prism	hexagon	8	18	12

A CUBOID IS A PRISM, AND SO IS A CUBE

Nothing new has to be learnt for them: they are the case where the cross-section happens to be a rectangle or a square.

The cylinder

A cylinder is a prism whose cross-section is a circle — except that its “rectangles” have curled into a single curved surface.



$$S = 2\pi r^2 + 2\pi rh$$

— two circles and one rectangle

The net of a cylinder — two circles and one rectangle

PART	SHAPE	ITS MEASUREMENT
the two ends	circles of radius r	πr^2 each
the curved surface	a rectangle when unrolled	$2\pi r \times h$
the whole surface	both together	$2\pi r^2 + 2\pi rh$

The unrolled rectangle is as long as the circle it came from — that is why its width is the circumference $2\pi r$.

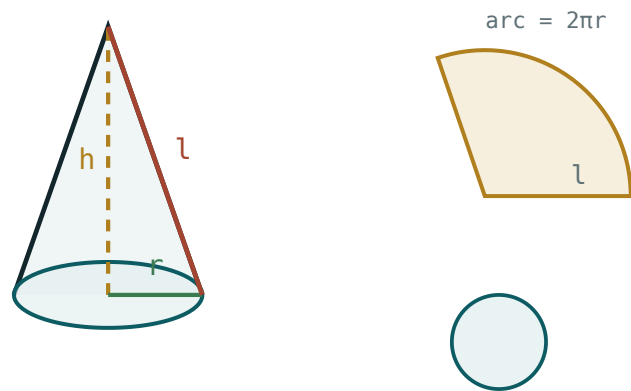
CYLINDER	END AREA	CURVED SURFACE	TOTAL SURFACE
$r = 5, h = 10$	78.5 cm^2	314 cm^2	471 cm^2
$r = 3, h = 8$	28.3 cm^2	150.7 cm^2	207.2 cm^2

TAKE $\pi \approx 3.14$ AND SAY SO

Every answer here is rounded. Writing “taking $\pi = 3.14$ ” beside the working is what makes 471 a correct answer rather than an approximate one.

Pyramids and the cone

A pyramid has one base and triangles rising to a single point, the **apex**. A cone is the same idea with a circular base and one curved surface.



$S = \pi r^2 + \pi r l$ – a circle and a sector
The net of a cone — a circle and a sector

SOLID	BASE	SIDES	FACES	EDGES	VERTICES
triangular pyramid	triangle	3 triangles	4	6	4
square pyramid	square	4 triangles	5	8	5
hexagonal pyramid	hexagon	6 triangles	7	12	7
cone	circle	one curved surface	1 flat	one curved edge	1 apex

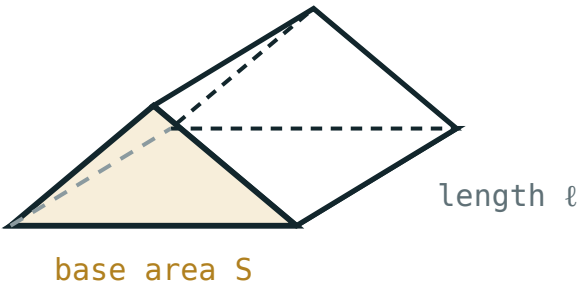
A PYRAMID HOLDS A THIRD OF THE PRISM ON THE SAME BASE

Pour sand from a pyramid into a prism of the same base and height and it fills exactly one third of it — and a cone fills a third of its cylinder. Grade 11 proves it; a jug of sand shows it today.

The volume of a prism

The cuboid rule $V = abc$ was really $(a \cdot b)$ — a base area — multiplied by the height. That is true of every prism, whatever its cross-section.

$V = \text{base area} \times \text{height}$



$V = S \cdot \ell$

Base area times height — the rule for every prism

PRISM	CROSS-SECTION AREA	LENGTH	VOLUME
triangular, base 6 by height 4	12 cm ²	10 cm	120 cm ³
trapezium ended	20 cm ²	8 cm	160 cm ³
cylinder, $r = 5$	78.5 cm ²	10 cm	785 cm ³
cylinder, $r = 3$	28.3 cm ²	8 cm	226 cm ³

ONE RULE, FOUR SOLIDS

Find the area of the end, multiply by the length. The only work that changes from one prism to the next is the area of the cross-section — a triangle, a trapezium, or a circle.

02 Worked examples

EXAMPLE 1 A triangular prism has a cross-section with base 6 cm and height 4 cm, and is 10 cm long. Find its volume.

① Cross-section $\frac{1}{2} \cdot 6 \cdot 4 = 12 \text{ cm}^2$.

② $12 \cdot 10$.

③ $= 120 \text{ cm}^3$.

Area in cm^2 times a length ✓

ANSWER

120 cm^3

EXAMPLE 2 A cylinder has radius 5 cm and height 10 cm. Find its volume, taking $\pi = 3.14$.

① End area $3.14 \cdot 25 = 78.5 \text{ cm}^2$.

② $78.5 \cdot 10$.

③ $= 785 \text{ cm}^3$.

Taking $\pi = 3.14$ ✓

ANSWER

785 cm^3

EXAMPLE 3 A prism has volume 240 cm^3 and length 8 cm . Find the area of its cross-section.

① $V = \text{area} \times 8.$

② $240 \div 8.$

③ $= 30 \text{ cm}^2.$
 $\text{cm}^3 \div \text{cm} = \text{cm}^2 \checkmark$

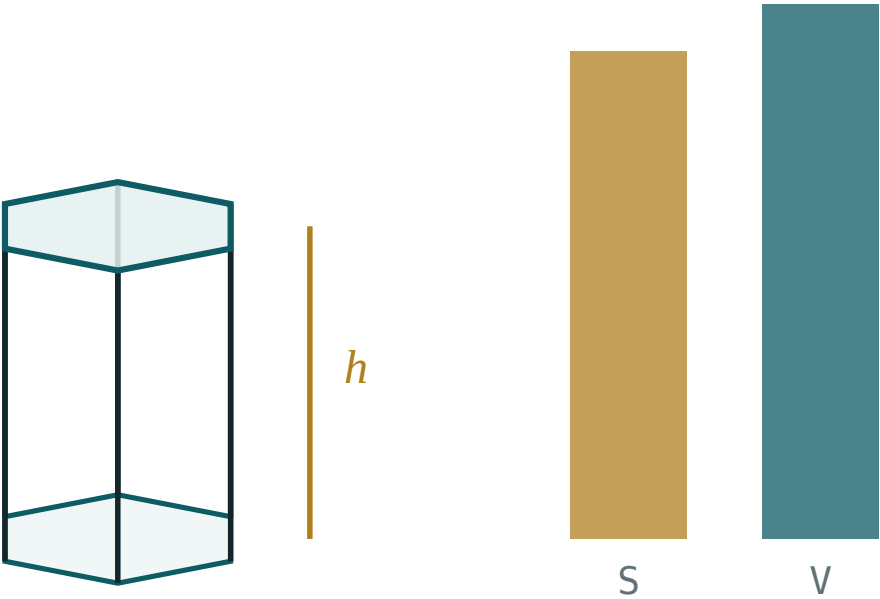
ANSWER

30 cm^2

03 Interactive model

Set the base of the model to four sides and then to many; as the polygon rounds off into a circle the prism becomes a cylinder in front of the class.

Base area times height



base area B **41.6**

perimeter P **24**

total S **227.1**

volume V **249.4**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Cross-section	Kesim	Поперечное сечение
Prism	Prizma	Призма
Cylinder	Silindr	Цилиндр
Cone	Konus	Конус
Pyramid	Piramida	Пирамида
Apex	Uchi (cho'qqi)	Вершина
Curved surface	Yon sirt	Боковая поверхность
Base area	Asos yuzasi	Площадь основания

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The cross-section of a prism is:

different at each cut

the same at every cut

always a circle

always a triangle

A cylinder has:

2 flat faces and 1 curved

3 flat faces

1 flat face

no faces

A cone has how many apexes?

0

1

2

4

The volume of any prism is:

base $\times$ height $\div$ 3

base area $\times$ height

base + height

base area $\div$ height

A cylinder of radius 5 and height 10 holds (taking $\pi = 3.14$):

157

314

785

78.5

The curved surface of a cylinder unrolls into:

a circle

a rectangle

a triangle

a sector

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The cross-section of a prism is

ANSWER The same all along

2. The faces of a cylinder

ANSWER 2 flat and 1 curved

3. The faces of a cone

ANSWER 1 flat and 1 curved

4. The volume of a prism

ANSWER Base area $\times$ height

5. A prism of cross-section 12 cm^2 and length 10 cm

ANSWER 120 cm^3

6. The cross-section of a cuboid

ANSWER A rectangle

7. The point at the top of a cone

ANSWER The apex

Medium · the standard question

7 PROBLEMS

1. A prism of cross-section 20 cm^2 and length 8 cm

ANSWER 160 cm^3

2. A cylinder of $r = 5 \text{ cm}$ and $h = 10 \text{ cm}$, $\pi = 3.14$

ANSWER 785 cm^3

3. A cylinder of $r = 3 \text{ cm}$ and $h = 8 \text{ cm}$

ANSWER 226 cm^3

4. The area of a circle of radius 5 cm

ANSWER 78.5 cm^2

5. The curved surface of a cylinder $r = 5$, $h = 10$

ANSWER 314 cm^2

6. The faces of a pyramid on a square base

ANSWER 5

7. The net of a cone

ANSWER A circle and a sector

Hard · stretch and reason

7 PROBLEMS

1. A cylinder of $r = 7$ cm and $h = 10$ cm

ANSWER 1539 cm^3

2. The total surface of a cylinder $r = 5$, $h = 10$

ANSWER 471 cm^2

3. A triangular prism, cross-section base 6 height 4, length 10

ANSWER 120 cm^3

4. A pyramid on the same base and height as a prism holds

ANSWER A third as much

5. A prism of volume 240 cm^3 and length 8 cm: its cross-section

ANSWER 30 cm^2

6. A cylinder of volume 628 cm^3 and radius 5 cm: its height

ANSWER 8 cm

7. Which solid has a circular cross-section all along?

ANSWER The cylinder

07 Homework

Homework — 5 tasks

Write “taking $\pi = 3.14$ ” beside every answer that uses it.

1. A prism has cross-section 15 cm^2 and length 12 cm. Find its volume.
2. A cylinder has radius 4 cm and height 10 cm. Find its volume.
3. Find the curved surface area of that cylinder.
4. Count the faces, edges and vertices of a pentagonal prism.
5. Name the cross-section of a cuboid, a cylinder and a triangular prism.

Building three-dimensional figures on an isometric grid

Drawing a solid honestly on flat paper — and reading it back from its three views.

LESSON 179–181

3 × 40 MIN

MATEMATIKA 6, §35

S7 8.4 3-D SHAPES

LESSON CLOCK

30'

30'

30'

25'

5

Drawing on the isometric grid	30 min
Counting the cubes	30 min
Plan and elevations	30 min
From views back to the solid	25 min
Homework	5 min

LEARNING OBJECTIVES

- Draw a solid made of unit cubes on an isometric grid.
- Count the cubes in a stack, including the hidden ones.
- Find the surface area of a stack of unit cubes.
- Draw the plan and the two elevations, and rebuild a solid from them.

TEXTBOOK

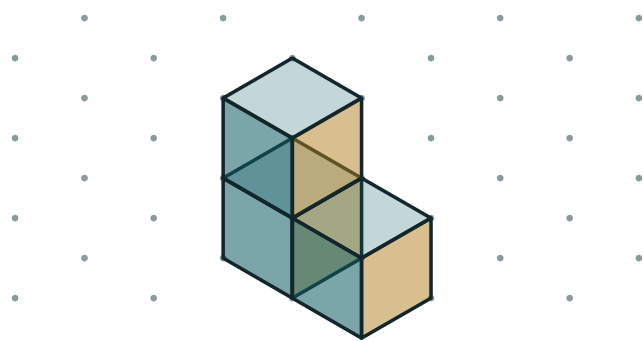
National	pp. 512–521
Cambridge	Stage 7, pp. 88–94
Workbook	Exercise 8.5

01

Explanation

Drawing on the isometric grid

Isometric paper carries dots in triangles rather than squares. Vertical edges are drawn vertically; the other two directions run along the sloping lines of the grid. Every edge of a unit cube is then the same length on the page.



three cubes on an isometric grid

Three unit cubes drawn on an isometric grid

RULE	WHY
vertical edges stay vertical	height must look like height
the other two directions follow the grid lines	they keep equal lengths equal
never draw a horizontal line	there are none in this projection
draw the front cubes last	they must cover what is behind them

THE DRAWING IS NOT A PHOTOGRAPH

In a real photograph the far edges would look shorter. On an isometric grid they do not, which is exactly what makes the drawing measurable — and slightly strange to look at.

Counting the cubes

Count layer by layer, from the bottom up. A cube you cannot see still has to be there if something is resting on it.

STACK	CUBES	HIDDEN	SURFACE, IN UNIT SQUARES
an L of three cubes	3	0	14
$2 \times 2 \times 1$	4	0	16
$3 \times 2 \times 2$	12	0	32
$2 \times 2 \times 2$	8	0	24
$3 \times 3 \times 3$	27	1	54

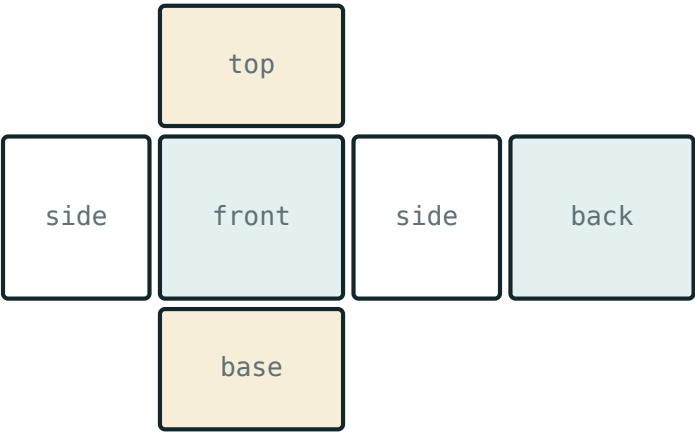
SURFACE AREA FROM THE JOINS

Three cubes have 18 faces in all. Two pairs are glued together, hiding 4 of them, so 14 are exposed. Every join hides exactly two faces — that is the quickest way to count.

Plan and elevations

Three flat drawings describe a solid without any perspective at all: the view from above and the views from the front and the side.

VIEW	LOOKING	SHOWS	FOR THE L OF THREE CUBES
plan	from above	length and depth	a 2 by 1 rectangle
front elevation	from the front	length and height	an L of three squares
side elevation	from the side	depth and height	a 1 by 2 rectangle



surface area = the area of the whole net

*Flat views and flat nets both turn a solid into drawings
on one page*

EACH VIEW IS A SHADOW, NOT A PICTURE

Anything behind something else disappears. That is why the plan of the L is only two squares, even though the solid has three cubes.

From views back to the solid

Given three views, rebuild the solid layer by layer: the plan gives the footprint, the elevations give the height of each part.

PLAN	FRONT	SIDE	SOLID
2×2	2×2	2×2	a $2 \times 2 \times 2$ cube, 8 cubes
3×1	3×1	1×1	a row of 3 cubes
2×1	an L	1×2	the L of 3 cubes
3×1	a staircase	1×3	a staircase of 6 cubes

THREE VIEWS DO NOT ALWAYS FIX THE NUMBER OF CUBES

A cube tucked out of sight behind another changes none of the views. The views give the shape from outside; only the drawing or the solid itself gives the count for certain.

02 Worked examples

EXAMPLE 1 A stack is 3 cubes long, 2 deep and 2 high. How many cubes, and what is its surface area in unit squares?

- 1 $3 \cdot 2 \cdot 2 = 12$ cubes.
- 2 The three faces are 6, 4 and 6.
- 3 $2(6 + 4 + 6) = 32$ unit squares.
It is a cuboid, so the old rule applies ✓

ANSWER

12 cubes, surface 32

EXAMPLE 2 Three cubes form an L: two in a row, one on top of the left-hand cube. Find the exposed surface.

- 1 $3 \cdot 6 = 18$ faces altogether.
- 2 Two joins, each hiding 2 faces: 4 hidden.
- 3 $18 - 4 = 14$ unit squares.
Count the joins, not the faces ✓

ANSWER

14 unit squares

EXAMPLE 3 Describe the plan and the two elevations of that L of three cubes.

- ① From above, the top cube hides the one below: a 2 by 1 rectangle.
- ② From the front, heights 2 and 1: an L of three squares.
- ③ From the side, one deep and two high: a 1 by 2 rectangle.
Each view is a shadow.

ANSWER

A 2×1 rectangle, an L, and a 1×2 rectangle

03 Interactive model

Build each solid from real cubes before drawing it; the hidden cube in a $3 \times 3 \times 3$ stack is only convincing when the class has had to put it there.

Cubes, views and surfaces

A $2 \times 2 \times 2$ stack contains:

A $3 \times 3 \times 3$ stack has how many cubes no face of which can be seen?

Three cubes joined in an L expose:

A $2 \times 2 \times 1$ block has surface:

12

16

20

24

The plan is the view from:

the front

the side

above

below

The side elevation shows:

length and depth

depth and height

length and height

nothing

On isometric paper you never draw:

a vertical line

a horizontal line

a sloping line

a dot

Three views fix the number of cubes:

- always
- never
- not always
- only for cuboids

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Isometric grid	Izometrik to‘r	Изометрическая сетка
Unit cube	Birlik kub	Единичный куб
Stack	Ustma-ust taxlam	Стопка
Plan	Ustdan ko‘rinish	Вид сверху
Front elevation	Oldindan ko‘rinish	Вид спереди
Side elevation	Yon ko‘rinish	Вид сбоку
Hidden cube	Yashirin kub	Скрытый куб
Exposed face	Ochiq yoq	Открытая грань

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

On an isometric grid the dots form:

squares

triangles

circles

hexagons

A vertical edge of a cube is drawn:

sloping left

sloping right

vertically

horizontally

Each join between two cubes hides:

1 face

2 faces

4 faces

nothing

A cube with something resting on it:

need not be there

must be there

is optional

is hollow

The front elevation shows:

length and depth

length and height

depth and height

volume

A view is:

a photograph

a shadow, with nothing behind

a net

a cross-section

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Cubes in a $2 \times 2 \times 1$ stack

ANSWER 4

2. Cubes in a $2 \times 2 \times 2$ stack

ANSWER 8

3. Cubes in a $3 \times 2 \times 2$ stack

ANSWER 12

4. Cubes in a $3 \times 3 \times 3$ stack

ANSWER 27

5. The dots of an isometric grid form

ANSWER Triangles

6. A vertical edge is drawn

ANSWER Vertically

7. The plan is the view from

ANSWER Above

Medium · the standard question

7 PROBLEMS

1. The surface of a $2 \times 2 \times 1$ block

ANSWER 16 unit squares

2. The surface of a $2 \times 2 \times 2$ block

ANSWER 24 unit squares

3. Cubes in a staircase of 1, 2 and 3

ANSWER 6

4. The exposed faces of an L of three cubes

ANSWER 14

5. The front elevation of a $2 \times 2 \times 2$ stack

ANSWER A 2 by 2 square

6. The plan of a $3 \times 2 \times 2$ stack

ANSWER A 3 by 2 rectangle

7. Cubes with no visible face in a $3 \times 3 \times 3$ stack

ANSWER 1

Hard · stretch and reason

7 PROBLEMS

1. The surface of a $3 \times 2 \times 2$ block

ANSWER 32 unit squares

2. The surface of a $3 \times 3 \times 3$ block

ANSWER 54 unit squares

3. The exposed faces of a staircase of 1, 2, 3 cubes

ANSWER 24

4. A plan, front and side all 2 by 2: the largest possible solid

ANSWER A $2 \times 2 \times 2$ cube of 8 cubes

5. Why can three views not fix the number of cubes?

ANSWER A hidden cube changes none of them

6. Which view shows depth and height?

ANSWER The side elevation

7. How many faces does each join between two cubes hide?

ANSWER 2

07 Homework

Homework — 5 tasks

Use isometric paper for the drawings and squared paper for the three views.

1. Draw a $2 \times 2 \times 2$ stack of cubes on isometric paper.
2. Draw the L of three cubes and write its plan, front and side elevations.
3. Find the number of cubes and the surface area of a $4 \times 2 \times 2$ stack.
4. Build a staircase of 1, 2, 3 and 4 cubes and count the cubes.
5. Draw a solid of your own from six cubes and give its three views.

Nets of three-dimensional figures

Unfold the solid, and its surface area becomes an ordinary area on flat paper.

LESSON 182–184

3 × 40 MIN

MATEMATIKA 6, §36

S7 8.4 3-D SHAPES

LESSON CLOCK

25'

30'

35'

25'

5

What a net is	25 min
The nets of a cube	30 min
Nets of prisms, pyramids and cones	35 min
Surface area from a net	25 min
Homework	5 min

LEARNING OBJECTIVES

- Draw a net of a cube, a cuboid and a prism.
- Decide whether a given arrangement of squares folds into a cube.
- Name the parts of the net of a cylinder, a cone and a pyramid.
- Find a surface area by adding the parts of the net.

TEXTBOOK

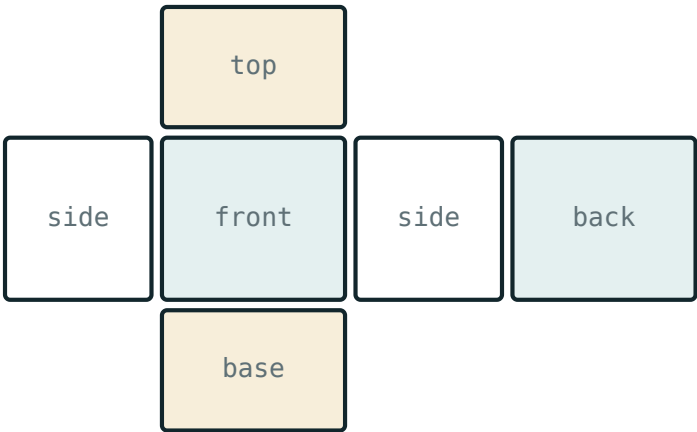
National	pp. 522–531
Cambridge	Stage 7, pp. 90–96
Workbook	Exercise 8.6

01

Explanation

What a net is

A net is the solid cut along some of its edges and laid flat, with every face still joined to at least one neighbour. Fold it up and the original solid comes back.



surface area = the area of the whole net

The net of a cuboid — six rectangles in three equal pairs

SOLID	ITS NET
cube	six equal squares
cuboid	six rectangles, in three equal pairs
triangular prism	two triangles and three rectangles
square pyramid	one square and four triangles
cylinder	two circles and one rectangle
cone	one circle and one sector

THE AREA OF THE NET IS THE SURFACE AREA OF THE SOLID

Nothing is lost or added by folding. Every surface-area question in this course could be answered by drawing the net and adding its pieces.

The nets of a cube

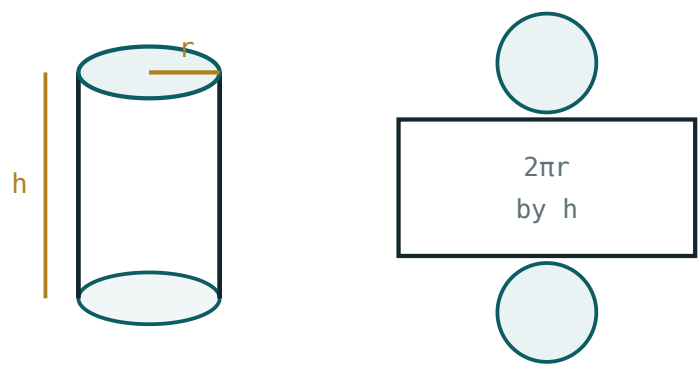
Six squares can be arranged in many ways, but only some of them fold into a cube.
There are exactly 11 genuinely different ones.

ARRANGEMENT	FOLDS INTO A CUBE?	WHY
a row of four with one square each side	yes	the classic cross
a row of three, two above, one below	yes	one of the eleven
a row of six	no	it wraps round and two faces land on the same place
a 2 by 3 rectangle	no	there is no square left for the top
four in a square with two attached	sometimes	it depends where the two are

THE TEST: WHICH FACES END UP OPPOSITE?

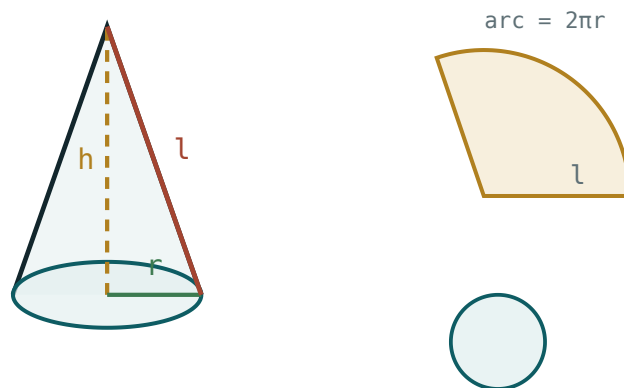
In a row of squares, the first and the fourth become opposite faces. If two squares of the net would have to become the *same* face, the arrangement fails. Cutting the shape out and folding it is the honest check — and the one worth doing.

Nets of prisms, pyramids and cones



$S = 2\pi r^2 + 2\pi rh$ – two circles and one rectangle
Two circles and a rectangle whose width is the circumference

SOLID	PIECES OF THE NET	THE MEASUREMENT THAT SURPRISES
prism on an n -sided base	2 ends and n rectangles	each rectangle is as long as the prism
cylinder	2 circles and 1 rectangle	the rectangle's width is $2\pi r$
pyramid on an n -sided base	1 base and n triangles	the triangle's height is the slant height, not the solid's
cone	1 circle and 1 sector	the sector's radius is the slant height



$$S = \pi r^2 + \pi r l - \text{a circle and a sector}$$

The net of a cone — a base circle and a sector

THE SLANT HEIGHT IS NOT THE HEIGHT

In a pyramid the triangle leans outwards, so its own height is longer than the height of the solid. Using the wrong one is the single commonest error in net questions.

Surface area from a net

SOLID	THE PIECES	SURFACE AREA
cube, edge 4 cm	6 squares of 16	96 cm^2
cuboid $5 \times 4 \times 3 \text{ cm}$	$2(20 + 12 + 15)$	94 cm^2
triangular prism, 3-4-5 ends, 10 cm long	$2 \cdot 6 + 30 + 40 + 50$	132 cm^2
cylinder, $r = 5$, $h = 10$	$2 \cdot 78.5 + 314$	471 cm^2
square pyramid, base 6, slant 5	$36 + 4 \cdot 15$	96 cm^2
cone, $r = 3$, slant 5	$28.3 + 47.1$	75.4 cm^2

LIST THE PIECES BEFORE ADDING ANYTHING

Write the shapes down — “two triangles of 6, three rectangles of 30, 40, 50” — and the addition can hardly go wrong. Most lost marks here are a forgotten face, not bad arithmetic.

02 Worked examples

EXAMPLE 1 Find the surface area of a $5 \times 4 \times 3$ cm cuboid from its net.

① The net has three pairs: 20, 12 and 15 cm².

② $2(20 + 12 + 15)$.

③ = 94 cm².

The same answer as the formula ✓

ANSWER

94 cm²

EXAMPLE 2 A triangular prism has 3-4-5 cm ends and is 10 cm long. Find its surface area.

① Two triangles of $\frac{1}{2} \cdot 3 \cdot 4 = 6$ cm².

② Three rectangles: $3 \cdot 10 = 30$, $4 \cdot 10 = 40$, $5 \cdot 10 = 50$.

③ $12 + 120 = 132$ cm².

Five pieces, all listed ✓

ANSWER

132 cm²

EXAMPLE
3**A cylinder has radius 5 cm and height 10 cm. Find its surface area from the net, taking $\pi = 3.14$.**

- ① Two circles of 78.5 cm^2 .
- ② A rectangle 31.4 cm by $10 \text{ cm} = 314 \text{ cm}^2$.
Its width is the circumference.
- ③ $157 + 314 = 471 \text{ cm}^2$.

ANSWER **471 cm^2** **03 Interactive model**

Give out six paper squares taped in a row and ask the class to make a cube; the failure is more instructive than any diagram of the eleven nets.

Folding and unfolding

A net of a cube has:

4 squares

6 squares

8 squares

11 squares

The number of genuinely different nets of a cube is:

6

8

11

24

A row of six squares folds into a cube:

yes

no

sometimes

only if bent

The net of a cylinder is:

two circles and a rectangle

a circle and a sector

six rectangles

two triangles

The rectangle in that net has width:

r

$2r$

πr

$2\pi r$

The net of a cone is:

a circle and a rectangle

a circle and a sector

two sectors

a triangle

The triangles of a pyramid's net use:

the height of the solid

the slant height

the base only

the volume

The area of a net equals:

the volume

the surface area

half the surface area

nothing useful

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Net	Yoyilma	Развёртка
To fold	Buklamoq	Складывать
Opposite faces	Qarama-qarshi yoqlar	Противоположные грани
Tab (flap)	Yopishtirish qismi	Клапан
Slant height	Yon balandlik (apofema)	Апофема
Sector	Sektor	Сектор
Circumference	Aylana uzunligi	Длина окружности
Surface area	Sirt yuzasi	Площадь поверхности

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A net is the solid:

cut in half

cut along edges and laid flat

drawn in perspective

filled with cubes

A cuboid's net has:

3 rectangles

6 rectangles in 3 pairs

6 squares

4 rectangles

A prism on a hexagonal base has a net with:

6 rectangles and 2 hexagons

6 triangles

8 squares

2 circles

A 2 by 3 rectangle of squares folds into a cube:

yes

no

sometimes

only with tabs

The surface area of a cube of edge 4 cm from its net is:

16

64

96

24

The commonest error in pyramid nets is using:

the base

the height instead of the slant height

too many triangles

the wrong unit

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The faces in the net of a cube

ANSWER 6

2. The number of different nets of a cube

ANSWER 11

3. The net of a cylinder

ANSWER Two circles and a rectangle

4. The net of a cone

ANSWER A circle and a sector

5. The net of a square pyramid

ANSWER A square and four triangles

6. The net of a triangular prism

ANSWER Two triangles and three rectangles

7. The area of a net equals

ANSWER The surface area of the solid

Medium · the standard question

7 PROBLEMS

1. The area of the net of a $5 \times 4 \times 3$ cm cuboid

ANSWER 94 cm^2

2. The area of the net of a cube of edge 4 cm

ANSWER 96 cm^2

3. The area of the net of a cube of edge 5 cm

ANSWER 150 cm^2

4. The width of the rectangle in a cylinder's net

ANSWER $2\pi r$

5. That width for $r = 5$ cm, $\pi = 3.14$

ANSWER 31.4 cm

6. The rectangles in the net of a hexagonal prism

ANSWER 6

7. The triangles in the net of a pentagonal pyramid

ANSWER 5

Hard · stretch and reason

7 PROBLEMS

1. The net of a triangular prism, 3-4-5 ends, 10 cm long

ANSWER 132 cm²

2. The net of a cylinder, $r = 5$, $h = 10$

ANSWER 471 cm²

3. The net of a square pyramid, base 6, slant 5

ANSWER 96 cm²

4. The net of a cone, $r = 3$, slant 5

ANSWER 75.4 cm²

5. Why does a row of six squares not fold into a cube?

ANSWER It wraps round and two faces meet in one place

6. The area of the rectangle in the net of a cylinder $r = 5$, $h = 10$

ANSWER 314 cm²

7. Which squares of a cube net must not touch?

ANSWER Those that will become opposite faces

07 Homework

Homework — 5 tasks

Draw the nets on squared paper, list every piece, then add.

1. Draw a net of a cube of edge 3 cm and find its area.
2. Draw a net of a $6 \times 4 \times 2$ cm cuboid and find its area.
3. A cylinder has radius 4 cm and height 9 cm. Sketch its net and give the dimensions of the rectangle.
4. Find the total surface area of that cylinder, taking $\pi = 3.14$.

5. Draw two different nets of a cube, and explain why a row of six squares is not one of them.

Revision — three-dimensional shapes

The families, the counts, the nets, the volumes and the three views, in one block.

LESSON 185–188

4 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 8 CONSOLIDATION

LESSON CLOCK

30'

35'

35'

35'

20'

5

The families and their counts	30 min
Nets and surface areas	35 min
Volumes	35 min
Views and drawings	35 min
The mistakes that cost marks	20 min
Homework	5 min

LEARNING OBJECTIVES

- Name any common solid and give its faces, edges and vertices.
- Draw or describe its net and find the surface area from it.
- Find the volume of any prism, including a cylinder.
- Move between an isometric drawing and the three views.

TEXTBOOK

National	pp. 532–541
Cambridge	Stage 7, pp. 84–96
Workbook	Review 8

01

Explanation

The families and their counts

SOLID	FACES	EDGES	VERTICES	$F + V - E$
cube or cuboid	6	12	8	2
triangular prism	5	9	6	2
hexagonal prism	8	18	12	2
square pyramid	5	8	5	2
hexagonal pyramid	7	12	7	2
prism on n sides	$n + 2$	$3n$	$2n$	2
pyramid on n sides	$n + 1$	$2n$	$n + 1$	2

THE CURVED SOLIDS STAND OUTSIDE THE PATTERN

A cylinder has 2 flat faces and 1 curved surface; a cone 1 and 1; a sphere 0 and 1. $F + V - E = 2$ applies to none of them.

Nets and surface areas

SOLID	NET	WORKED EXAMPLE	SURFACE AREA
cube	6 squares	edge 4 cm	96 cm^2
cuboid	3 pairs of rectangles	$5 \times 4 \times 3 \text{ cm}$	94 cm^2
triangular prism	2 triangles, 3 rectangles	3-4-5 ends, 10 cm long	132 cm^2
cylinder	2 circles, 1 rectangle	$r = 5, h = 10$	471 cm^2
square pyramid	1 square, 4 triangles	base 6, slant 5	96 cm^2
cone	1 circle, 1 sector	$r = 3, \text{slant } 5$	75.4 cm^2

TWO MEASUREMENTS ARE EASY TO CONFUSE

In a cylinder the rectangle's width is $2\pi r$, not r . In a pyramid the triangles use the slant height, not the height of the solid. Almost every lost mark in this block is one of those two.

Volumes

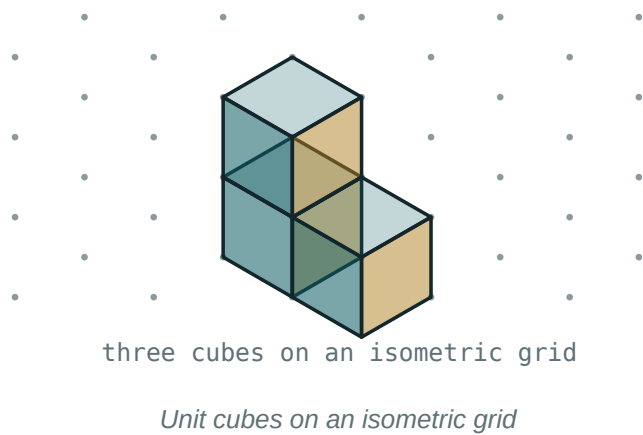
$V = \text{base area} \times \text{height}$ for every prism, including the cylinder

SOLID	BASE AREA	HEIGHT OR LENGTH	VOLUME
$5 \times 4 \times 3 \text{ cm cuboid}$	20 cm^2	3 cm	60 cm^3
cube of edge 4 cm	16 cm^2	4 cm	64 cm^3
triangular prism	12 cm^2	10 cm	120 cm^3
cylinder $r = 5$	78.5 cm^2	10 cm	785 cm^3
cylinder $r = 3$	28.3 cm^2	8 cm	226 cm^3

REVERSE IT THE SAME WAY AS BEFORE

Cross-section = $V \div \text{length}$; length = $V \div \text{cross-section}$. A prism of 240 cm^3 and length 8 cm has a cross-section of 30 cm^2 .

Views and drawings



STACK	CUBES	SURFACE	PLAN
L of three cubes	3	14	2×1
$2 \times 2 \times 1$	4	16	2×2
$2 \times 2 \times 2$	8	24	2×2
$3 \times 2 \times 2$	12	32	3×2
$3 \times 3 \times 3$	27	54	3×3

THREE VIEWS, THREE PAIRS OF DIMENSIONS

The plan gives length and depth, the front elevation length and height, the side elevation depth and height. Each dimension appears in two of the three views, which is how the drawings check one another.

The mistakes that cost marks

MISTAKE	LOOKS LIKE	CORRECT
hidden edges not counted	<i>16 edges on a hexagonal prism</i>	18
the cylinder's rectangle given width r	5×10	31.4×10
the pyramid's height used for its triangles	$4 \cdot \frac{1}{2} \cdot 6 \cdot 4$	$4 \cdot \frac{1}{2} \cdot 6 \cdot 5$
a face of the net forgotten	<i>5 faces of a cuboid</i>	6
surface area written in cm^3	$94\ cm^3$	$94\ cm^2$

THREE OF THE FIVE ARE A MISSING PIECE

An edge, a face, or the second measurement of a rectangle. Listing every part before adding is the whole defence.

02 Worked examples

EXAMPLE 1 A hexagonal prism: find its faces, edges and vertices, and check them.

1 $n = 6$, so $F = 8$, $E = 18$, $V = 12$.

2 $F + V - E = 8 + 12 - 18$.

3 $= 2 \checkmark$

The check catches a missed upright edge.

ANSWER

8 faces, 18 edges, 12 vertices

EXAMPLE 2 A cylinder has radius 5 cm and height 10 cm. Find its volume and its surface area, taking $\pi = 3.14$.

- 1 Base area 78.5 cm^2 , so $V = 785 \text{ cm}^3$.
- 2 Two circles 157 cm^2 , rectangle $31.4 \cdot 10 = 314 \text{ cm}^2$.
- 3 Surface 471 cm^2 .
Volume in cm^3 , area in cm^2 ✓

ANSWER

785 cm^3 and 471 cm^2

EXAMPLE 3 A prism has volume 240 cm^3 and length 8 cm. Find the area of its cross-section and describe its plan if the cross-section is a 5 by 6 rectangle.

- 1 $240 \div 8 = 30 \text{ cm}^2$.
- 2 A 5 by 6 rectangle has area 30 ✓
- 3 Standing on that face, the plan is 5 by 6.
The prism is a $5 \times 6 \times 8$ cuboid.

ANSWER

30 cm^2 ; the plan is a 5 by 6 rectangle

03 Interactive model

Work the revision from the mistake table backwards: give the wrong answers first and let the class find each error. It fixes them faster than doing the questions correctly once.

The solids block in eight questions

A hexagonal prism has how many edges?

A square pyramid has how many faces?

The volume of any prism is:

A cylinder $r = 5$, $h = 10$ holds:

157

314

471

785

Its surface area is:

157

314

471

785

A $3 \times 3 \times 3$ stack of unit cubes has surface:

27

36

54

81

The plan of a solid shows:

length and height

length and depth

depth and height

the volume

$F + V - E$ for a cone is:

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Prism	Prizma	Призма
Pyramid	Piramida	Пирамида
Cylinder	Silindr	Цилиндр
Cone	Konus	Конус
Net	Yoyilma	Развёртка
Cross-section	Kesim	Поперечное сечение
Elevation	Ko'rinish	Вид
Slant height	Yon balandlik	Апофема

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A prism on an n -sided base has faces:

n

$n + 1$

$n + 2$

$2n$

A pyramid on an n -sided base has edges:

n

$n + 1$

$2n$

$3n$

The cylinder's net rectangle is as wide as:

r

$2r$

πr

$2\pi r$

The triangles of a pyramid net use:

the height

the slant height

the base

the volume

Surface area is measured in:

cm

cm^2

cm^3

litres

A count giving $F + V - E = 3$ means:

an unusual solid

a wrong count

a curved surface

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Faces, edges and vertices of a cube

ANSWER 6, 12, 8

2. Faces, edges and vertices of a triangular prism

ANSWER 5, 9, 6

3. $F + V - E$ for any flat-faced solid

ANSWER 2

4. The net of a cylinder

ANSWER Two circles and a rectangle

5. The volume of a prism

ANSWER Base area $\times$ height

6. The plan is the view from

ANSWER Above

7. The number of different nets of a cube

ANSWER 11

Medium · the standard question

7 PROBLEMS

1. Faces, edges and vertices of a hexagonal prism

ANSWER 8, 18, 12

2. Faces, edges and vertices of a square pyramid

ANSWER 5, 8, 5

3. The volume of a $5 \times 4 \times 3$ cm cuboid

ANSWER 60 cm^3

4. Its surface area

ANSWER 94 cm^2

5. A prism of cross-section 12 cm^2 , length 10 cm

ANSWER 120 cm^3

6. A cylinder $r = 5$, $h = 10$, $\pi = 3.14$

ANSWER 785 cm^3

7. The surface of a $2 \times 2 \times 2$ block of unit cubes

ANSWER 24

Hard · stretch and reason

7 PROBLEMS

1. The surface area of a cylinder $r = 5$, $h = 10$

ANSWER 471 cm^2

2. The surface area of a 3-4-5 triangular prism 10 cm long

ANSWER 132 cm^2

3. The surface area of a square pyramid, base 6, slant 5

ANSWER 96 cm^2

4. Cubes in a $3 \times 3 \times 3$ stack, and how many are hidden

ANSWER 27 and 1

5. A prism of volume 240 cm^3 and length 8 cm: its cross-section

ANSWER 30 cm^2

6. The edges of a prism on an n -sided base

ANSWER $3n$

7. Why can three views not fix the number of cubes?

ANSWER A hidden cube changes none of them

07 Homework

Homework — 5 tasks

List every face of a net before adding, and check every set of counts with $F + V - E = 2$.

1. Give the faces, edges and vertices of a pentagonal prism and of a pentagonal pyramid.
2. Find the surface area of a cuboid 7 cm by 5 cm by 3 cm from its net.
3. A cylinder has radius 3 cm and height 10 cm. Find its volume and surface area.
4. A triangular prism has a cross-section of area 18 cm^2 and is 9 cm long. Find its volume.
5. Draw a $3 \times 2 \times 1$ stack of cubes and give its plan and two elevations.

Control work 10 — three-dimensional shapes

Counts, nets, surface areas, volumes and views, in twenty marks.

LESSON 189

1 × 40 MIN

MATEMATIKA 6, NAZORAT ISHI 10

S7 8 REVIEW

LESSON CLOCK

2'

28'

8'

2'

Instructions

2 min

The paper

28 min

Answers and diagnosis

8 min

What comes next

2 min

LEARNING OBJECTIVES

- Give the faces, edges and vertices of a named solid.
- Find a surface area and a volume for a cuboid, a prism and a cylinder.
- Describe the net of a cylinder with its measurements.
- Name each lost mark by its type and rewrite the whole solution.

TEXTBOOK

National

pp. 498–541

Cambridge

Stage 7, pp. 84–96

Workbook

Control paper 10

01

Explanation

The paper — 20 marks, 28 minutes

Q	TASK	MARKS	FROM
1	The faces, edges and vertices of a hexagonal prism	3	L175
2	The faces, edges and vertices of a square pyramid	2	L175
3	The surface area of a $6 \times 5 \times 4$ cm cuboid	3	L182–184
4	The volume of that cuboid	2	L176–178
5	A prism of cross-section 15 cm^2 and length 8 cm: its volume	2	L176–178
6	A cylinder of radius 4 cm and height 10 cm: its volume, $\pi = 3.14$	3	L176–178
7	Name the parts of a cylinder's net and give the rectangle's width for $r = 4$	3	L182–184
8	The number of cubes and the surface of a $2 \times 2 \times 2$ stack	2	L179–181

THE ANSWERS

8, 18, 12; 5, 8, 5; 148 cm²; 120 cm³; 120 cm³; 502.4 cm³; two circles and a rectangle of width 25.12 cm; 8 cubes and 24 unit squares.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
hidden edges not counted	16 edges	18
the base of a pyramid forgotten	4 faces	5
surface area written in cm ³	148 cm ³	148 cm ²
a prism's parts added, not multiplied	15 + 8	15 · 8
π left out of the cylinder	16 · 10 = 160	3.14 · 16 · 10
the net's rectangle given width r	4 cm	2πr = 25.12 cm
every cube counted with six faces	48	24

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

What comes next

IF YOU LOST MARKS ON	REVISE
Q1–Q2	counting faces, edges and vertices, L175
Q3 and Q7	nets and surface areas, L182–184
Q4–Q6	volumes of prisms and cylinders, L176–178
Q8	stacks of cubes and their views, L179–181

LOOKING FORWARD

One more Think task on nets, then the general revision of the whole Grade 6 course, and finally the presentation of the survey project.

02 Worked examples

EXAMPLE 1 Model answer, Q3: the surface area of a $6 \times 5 \times 4$ cm cuboid.

① The three faces: 30, 20 and 24 cm².

② $2(30 + 20 + 24)$.

③ $= 148$ cm².
Area, so cm² ✓

ANSWER

148 cm²

EXAMPLE 2 Model answer, Q6: a cylinder of radius 4 cm and height 10 cm.

① Base area $3.14 \cdot 16 = 50.24$ cm².
Taking $\pi = 3.14$.

② $50.24 \cdot 10$.

③ $= 502.4$ cm³.

ANSWER

502.4 cm³

EXAMPLE 3 Model answer, Q7: the net of a cylinder of radius 4 cm.

- ① Two circles of radius 4 cm and one rectangle.
- ② The rectangle is as wide as the circumference, $2\pi r$.
- ③ $2 \cdot 3.14 \cdot 4 = 25.12$ cm.
Not 4 cm.

ANSWER

Two circles and a rectangle 25.12 cm wide

03 Interactive model

Return Q7 first and ask who wrote 4 cm; naming it as “the width is the circumference” settles it for the year.

The paper in eight questions

A hexagonal prism has edges:

A square pyramid has faces:

The surface area of a $6 \times 5 \times 4$ cuboid is:

Its volume is:

74

120

148

15

A prism of cross-section 15 cm^2 and length 8 cm holds:

23

30

120

60

A cylinder $r = 4$, $h = 10$ holds:

160

502.4

251.2

125.6

The rectangle in its net is wide:

4 cm

12.56 cm

25.12 cm

50.24 cm

A $2 \times 2 \times 2$ stack of unit cubes has surface:

8

16

24

48

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	Ў‘ЗБЕКЧА	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Prism	Prizma	Призма
Pyramid	Piramida	Пирамида
Cylinder	Silindr	Цилиндр
Net	Yoyilma	Развёртка
Surface area	Sirt yuzasi	Площадь поверхности
Volume	Hajm	Объём
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 is easiest to get wrong on:

the faces

the edges

the vertices

the name

Q3 wants an answer in:

cm

cm^2

cm^3

litres

Q5 multiplies the cross-section by:

three

the length

the surface

six

Q6 needs:

no π

π once

π twice

a square root

Q7 asks for the rectangle's width, which is:

the radius

the diameter

the circumference

the height

Q8 counts each glued pair of faces as:

visible

hidden

double

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Faces, edges and vertices of a hexagonal prism

ANSWER 8, 18, 12

2. Faces, edges and vertices of a square pyramid

ANSWER 5, 8, 5

3. The volume of a $6 \times 5 \times 4$ cm cuboid

ANSWER 120 cm^3

4. Its surface area

ANSWER 148 cm^2

5. A prism of cross-section 15 cm^2 , length 8 cm

ANSWER 120 cm^3

6. The parts of a cylinder's net

ANSWER Two circles and a rectangle

7. Cubes in a $2 \times 2 \times 2$ stack

ANSWER 8

Medium · the standard question

7 PROBLEMS

1. A cylinder $r = 4$ cm, $h = 10$ cm, $\pi = 3.14$

ANSWER 502.4 cm³

2. The width of the rectangle in that net

ANSWER 25.12 cm

3. The surface of a $2 \times 2 \times 2$ stack of unit cubes

ANSWER 24

4. The base area of that cylinder

ANSWER 50.24 cm²

5. The curved surface of that cylinder

ANSWER 251.2 cm²

6. Its total surface area

ANSWER 351.68 cm²

7. $F + V - E$ for the hexagonal prism

ANSWER 2

Hard · stretch and reason

7 PROBLEMS

1. Why is 16 the wrong number of edges for a hexagonal prism?

ANSWER Two upright edges were hidden

2. Why is 160 cm^3 wrong for the cylinder?

ANSWER π was left out of the base area

3. Why is 48 wrong for the $2 \times 2 \times 2$ stack?

ANSWER The glued faces were counted

4. A prism of volume 120 cm^3 and length 8 cm: its cross-section

ANSWER 15 cm^2

5. A cuboid of surface 148 cm^2 , 6 by 5: the third edge

ANSWER 4 cm

6. A cylinder of volume 502.4 cm^3 and radius 4 cm: its height

ANSWER 10 cm

7. The commonest lost mark in this paper

ANSWER The width of the cylinder's rectangle

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, with the units on every line.

1. Rewrite in full every question on which you lost a mark, naming the slip in the margin.
2. Give the faces, edges and vertices of an octagonal prism and check them.
3. Find the volume and the surface area of a cuboid 8 cm by 4 cm by 3 cm.
4. A cylinder has radius 5 cm and height 6 cm. Find its volume and its total surface area.
5. Draw a $3 \times 2 \times 1$ stack of unit cubes and give its surface area in unit squares.

Think — nets in context

Designing a real box: the same six faces, laid out two ways, and two very different bills for card.

LESSON 190

1 × 40 MIN

MATEMATIKA 6, O'YLAB KO'R

S7 8.4 IN CONTEXT

LESSON CLOCK

10'

12'

12'

6'

The task	10 min
Two layouts for the same box	12 min
Waste, tabs and the sheet	12 min
Back to the shape of the box	6 min

LEARNING OBJECTIVES

- Design a net for a box of given dimensions, with tabs.
- Find the smallest rectangle of card the net will fit into.
- Compare two layouts by the card each wastes.
- Relate the result back to the shape of the box itself.

TEXTBOOK

National	pp. 542–544
Cambridge	Stage 7, pp. 94–96
Workbook	Investigation 8

01

Explanation

The task

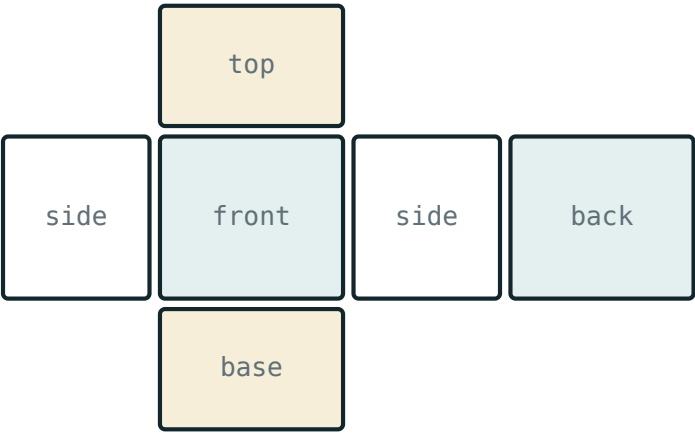
TODAY'S TASK

A box must hold 10 cm by 8 cm by 5 cm of tea. Design its net, decide how it should be laid out on a sheet of card, and say how much card is actually bought for each box.

QUANTITY	WORKING	VALUE
volume	$10 \cdot 8 \cdot 5$	400 cm^3
the three faces	$80, 40, 50$	cm^2
surface area	$2(80 + 40 + 50)$	340 cm^2

The 340 cm² is the card that ends up as box. It is not the card that is bought.

Two layouts for the same box



surface area = the area of the whole net

A net laid out flat — the pieces are fixed, their arrangement is not

LAYOUT	ARRANGEMENT	SMALLEST RECTANGLE	ITS AREA
the cross	a column of four faces with the two sides beside the front	$26 \times 26\text{ cm}$	676 cm^2
the row	four faces in a row, top above and bottom below	$36 \times 21\text{ cm}$	756 cm^2

WHERE THE 26S COME FROM

Down the column: $8 + 5 + 8 + 5 = 26\text{ cm}$. Across: $8 + 10 + 8 = 26\text{ cm}$. The row layout is $10 + 8 + 10 + 8 = 36\text{ cm}$ across and $8 + 5 + 8 = 21\text{ cm}$ down.

Waste, tabs and the sheet

LAYOUT	CARD BOUGHT	CARD USED	WASTE	WASTE AS A SHARE
the cross	676 cm^2	340 cm^2	336 cm^2	about 50%
the row	756 cm^2	340 cm^2	416 cm^2	about 55%

Tabs are extra. Seven glued edges of total length 46 cm, each given a 1 cm tab, add 46 cm² more card — and they must fit inside the same rectangle.

THE SURFACE AREA IS THE SAME IN BOTH LAYOUTS

Both boxes use 340 cm² of card as box. What differs is the rectangle it has to be cut from, and the offcuts are paid for. On a run of 10 000 boxes the cross layout saves 80 cm² each — 80 m² of card.

Back to the shape of the box

Project 3 of Quarter III found that, for a fixed volume, the cube uses least card. This box holds 400 cm^3 , so the cube would have edge 7.4 cm .

BOX	VOLUME	SURFACE AREA
$10 \times 8 \times 5$	400 cm^3	340 cm^2
cube of edge 7.4	<i>about 400 cm^3</i>	<i>about 326 cm^2</i>

TWO SAVINGS, OF VERY DIFFERENT SIZES

Reshaping the box towards a cube saves about 14 cm^2 ; laying the net out well saves 80 cm^2 . For this box the layout matters more than the shape — which is not what most people guess, and is exactly the kind of thing a report should say.

02

Worked examples

EXAMPLE 1 Find the card used, and the card bought, for the cross layout.

1

Used: $2(80 + 40 + 50) = 340\text{ cm}^2$.

2

Bought: the rectangle 26 by 26 cm .

3

 676 cm^2 , so 336 cm^2 is waste.
About half of what is bought ✓

ANSWER

 340 used, 676 bought

EXAMPLE 2 How much card does the cross layout save over the row layout on 10 000 boxes?

① $756 - 676 = 80 \text{ cm}^2$ a box.

② $80 \cdot 10\,000 = 800\,000 \text{ cm}^2$.

③ $800\,000 \div 10\,000 = 80 \text{ m}^2$.
 1 m^2 is $10\,000 \text{ cm}^2$.

ANSWER 80 m^2 **EXAMPLE 3** A cube holding the same 400 cm^3 has edge about 7.4 cm. Find its surface area and compare.

① $7.4^2 = 54.76 \text{ cm}^2$ a face.

② $6 \cdot 54.76 \approx 326 \text{ cm}^2$.

③ $340 - 326 = 14 \text{ cm}^2$ saved.
 Less than the 80 cm^2 the layout saves.

ANSWERAbout 326 cm^2 — a saving of 14 cm^2 **03 Interactive model**

Let pairs cut both layouts from squared paper and measure the offcuts; the “obvious” row layout losing to the cross is the moment worth waiting for.

Card used and card bought

A $10 \times 8 \times 5$ cm box has surface area:

170

340

400

676

Its volume is:

170

340

400

676

The cross layout fits into a rectangle:

26×26

36×21

34×20

26×36

The row layout fits into:

26×26

36×21

21×21

36×26

The cross layout wastes:

80

336

416

676

The cross saves, per box:

14 cm^2

80 cm^2

336 cm^2

nothing

A cube of the same volume saves about:

14 cm^2

80 cm^2

340 cm^2

nothing

For this box the bigger saving comes from:

the shape

the layout

the tabs

the colour

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Layout	Joylashtirish	Раскладка
Tab (flap)	Yopishtirish qismi	Клапан
Waste	Chiqindi	Отходы
Offcut	Qirqindi	Обрезок
Bounding rectangle	O'rovchi to'rtburchak	Охватывающий прямоугольник
To nest	Zich joylashtirmoq	Компоновать
Efficiency	Samaradorlik	Эффективность
Production run	Ishlab chiqarish partiyasi	Партия

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The card used as box is:

the bounding rectangle

the surface area

the volume

the waste

The card bought is:

the surface area

the smallest rectangle the net fits into

the volume

the tabs

Changing the layout changes:

the surface area

the waste

the volume

the number of faces

Tabs are needed so that:

the box looks better

the faces can be glued

the volume grows

the net is square

Nesting the nets of many boxes:

wastes more

shares the offcuts

changes the volume

needs no tabs

A good report gives:

only the best answer

the comparison and its limits

the volume alone

a drawing only

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The surface area of a $10 \times 8 \times 5$ cm box

ANSWER 340 cm^2

2. Its volume

ANSWER 400 cm^3

3. The faces in its net

ANSWER 6

4. Tabs are needed so that

ANSWER The faces can be glued together

5. The bounding rectangle of the cross layout

ANSWER 26 by 26 cm

6. Its area

ANSWER 676 cm^2

7. The card that ends up as box

ANSWER 340 cm^2

Medium · the standard question

7 PROBLEMS

1. The waste in the cross layout

ANSWER 336 cm²

2. The bounding rectangle of the row layout

ANSWER 36 by 21 cm

3. Its area

ANSWER 756 cm²

4. The waste in the row layout

ANSWER 416 cm²

5. Which layout wastes less

ANSWER The cross

6. The waste of the cross layout as a share

ANSWER About 50%

7. The waste of the row layout as a share

ANSWER About 55%

Hard · stretch and reason

7 PROBLEMS

1. A cube holding 400 cm^3 has edge, to 1 d.p.

ANSWER 7.4 cm

2. Its surface area

ANSWER About 326 cm^2

3. The card a cube would save

ANSWER About 14 cm^2 a box

4. The card the better layout saves

ANSWER 80 cm^2 a box

5. On 10 000 boxes, that saving

ANSWER 80 m^2

6. 1 cm tabs on seven edges totalling 46 cm add

ANSWER 46 cm^2

7. Why does the layout matter if the surface area is fixed?

ANSWER The offcuts of the rectangle are paid for too

07 Homework

Homework — the task

One page: the net with its measurements, the two layouts, the waste of each, and one conclusion.

1. Draw a net for a 12 cm by 6 cm by 4 cm box and find its surface area.
2. Lay that net out two different ways and give the smallest rectangle each fits into.
3. Work out the waste of each layout, in cm^2 and as a percentage.
4. Find the edge and the surface area of a cube holding the same volume.
5. Write one sentence saying whether the shape or the layout saves more card for your box.

General revision I — number and algebra

Directed numbers, fractions, expressions and equations, place value, roots and sequences.

LESSON 191–194

4 × 40 MIN

MATEMATIKA 6, YAKUNIY TAKRORLASH

S7 WHOLE-COURSE REVIEW

LESSON CLOCK

30'

35'

35'

35'

20'

5

Directed numbers	30 min
Fractions	35 min
Expressions and equations	35 min
Place value, decimals and roots	35 min
Sequences	20 min
Homework	5 min

LEARNING OBJECTIVES

- Calculate with directed numbers in all four operations.
- Divide fractions and add or subtract mixed numbers.
- Simplify an expression and solve a linear equation.
- Work with powers of ten, rounding, roots and the nth term.

TEXTBOOK

National	pp. 545–556
Cambridge	Stage 7, pp. 8–120
Workbook	Checkpoint practice 1

01

Explanation

Directed numbers

CALCULATION	RULE	ANSWER
$-7 + 3$	move right from -7	-4
$-5 - 4$	move left from -5	-9
$3 - (-8)$	subtracting a negative adds	11
$(-6)(-4)$	two negatives multiply to a positive	24
$-20 \div 5$	one negative gives a negative	-4

TWO RULES, NOT MANY

For adding and subtracting, think of movement on the number line. For multiplying and dividing, count the negative signs: an even number of them gives a positive answer, an odd number a negative.

Fractions

CALCULATION	METHOD	ANSWER
$\frac{3}{4} \div 2$	multiply by $\frac{1}{2}$	$\frac{3}{8}$
$5 \div \frac{1}{4}$	how many quarters in 5	20
$\frac{2}{3} \div \frac{4}{9}$	multiply by the reciprocal	$1\frac{1}{2}$
$2\frac{3}{4} + 1\frac{5}{6}$	wholes, then twelfths	$4\frac{7}{12}$
$4\frac{1}{3} - 1\frac{5}{6}$	regroup one whole	$2\frac{1}{2}$

A MIXED NUMBER IS CONVERTED BEFORE DIVIDING, NOT AFTER

$2\frac{1}{4} \div 1\frac{1}{2}$ is $\frac{9}{4} \div \frac{3}{2} = \frac{9}{4} \cdot \frac{2}{3} = \frac{3}{2}$. Turning over a mixed number as it stands gives nonsense.

Expressions and equations

TASK	WORKING	ANSWER
simplify $3x + 5x - 2x$	collect like terms	$6x$
expand $2(3x - 4)$	multiply both terms	$6x - 8$
evaluate $4x - 7$ at $x = 3$	$12 - 7$	5
solve $3x + 7 = 22$	$3x = 15$	$x = 5$
solve $5x - 4 = 3x + 10$	$2x = 14$	$x = 7$
solve $2(x - 3) = 14$	$x - 3 = 7$	$x = 10$

EVERY SOLUTION CAN BE CHECKED IN ONE LINE

Put $x = 7$ back into $5x - 4 = 3x + 10$: $31 = 31 \checkmark$. There is no excuse for a wrong answer to an equation surviving to the end of a paper.

Place value, decimals and roots

TASK	ANSWER	TASK	ANSWER
3.6×10^3	3600	$\sqrt{144}$	12
$4500 \div 10^2$	45	$\sqrt{81}$	9
0.05×10^4	500	$\sqrt[3]{64}$	4
$0.6 + 0.35$	0.95	$\sqrt[3]{125}$	5
2.4×0.5	1.2	$HCF(12, 18)$	6
$6.3 \div 0.7$	9	$LCM(12, 18)$	36
0.8×0.04	0.032	3.847 to 2 d.p.	3.85

COUNT THE DECIMAL PLACES IN A MULTIPLICATION

0.8 has one and 0.04 has two, so the answer has three: 0.032. In a division, make the divisor whole first — $6.3 \div 0.7$ becomes $63 \div 7$.

Sequences

SEQUENCE	DIFFERENCE	NTH TERM	THE 100TH
5, 8, 11, 14	3	$3n + 2$	302
3, 7, 11, 15	4	$4n - 1$	399
2, 9, 16, 23	7	$7n - 5$	695
20, 17, 14, 11	-3	$23 - 3n$	-277

WRITE DN, THEN CORRECT IT

The difference gives the dn ; the first term says what to add or subtract. Test the rule on the second term before using it — five seconds that saves the whole question.

02 Worked examples

EXAMPLE 1 Work out $3 - (-8) + (-6)(-4) \div 8$.

- ① $3 - (-8) = 11$.
Subtracting a negative adds.

- ② $(-6)(-4) = 24$, and $24 \div 8 = 3$.
Two negatives make a positive.

- ③ $11 + 3 = 14$.
Multiplication before addition ✓

ANSWER

14

EXAMPLE 2 Solve $5x - 4 = 3x + 10$ and check the answer.

- ① Subtract $3x$: $2x - 4 = 10$.

- ② Add 4: $2x = 14$, so $x = 7$.

- ③ Check: $35 - 4 = 31$ and $21 + 10 = 31$ ✓

ANSWER

$x = 7$

EXAMPLE 3 Find the n th term of 2, 9, 16, 23, ... and its 100th term.

- ① $d = 7$, so start from $7n$.
- ② $7n$ gives 7 at $n = 1$, but the term is 2: subtract 5.
- ③ $7n - 5$; at $n = 100$, 695.
Check $n = 2$: 9 ✓

ANSWER

$7n - 5$ and 695

03 Interactive model

Run the revision as a relay: one section per group, each producing three questions and their answers for the others. Writing the questions revises more than answering them.

Number and algebra in eight questions

$-5 - 4$ equals:

-1

-9

1

9

$(-6)(-4)$ equals:

-24

24

-10

10

$\frac{2}{3} \div \frac{4}{9}$ equals:

$\frac{8}{27}$

$1\frac{1}{2}$

$\frac{2}{3}$

6

$2\frac{3}{4} + 1\frac{5}{6}$ equals:

$$3\frac{8}{10}$$

$$4\frac{7}{12}$$

$$4\frac{1}{2}$$

$$3\frac{19}{12}$$

$2(3x - 4)$ equals:

$$6x - 4$$

$$6x - 8$$

$$5x - 8$$

$$6x + 8$$

$3x + 7 = 22$ gives:

$$x = 5$$

$$x = 7$$

$$x = 15$$

$$x = 9$$

$6.3 \div 0.7$ equals:

0.9

9

90

0.09

The n th term of $5, 8, 11, 14$ is:

$3n$

$3n + 2$

$5n$

$n + 3$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Directed number	Musbat va manfiy son	Целое число со знаком
Reciprocal	Teskari son	Обратное число
Like terms	O'xshash hadlar	Подобные слагаемые
Equation	Tenglama	Уравнение
Lowest common multiple	EKUK	НОК
Highest common factor	EKUB	НОД
Square root	Kvadrat ildiz	Квадратный корень
nth term	n-chi had	n-й член

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Subtracting a negative:

subtracts twice

adds

gives zero

changes nothing

Dividing by a fraction is the same as:

multiplying by it

multiplying by its reciprocal

adding it

subtracting it

Like terms can be:

multiplied only

collected

ignored

squared

A solution should always be:

rounded

checked in the original equation

doubled

left as a fraction

0.8×0.04 has how many decimal places?

1

2

3

4

In $dn + (a - d)$ the d is:

the first term

the common difference

the position

the answer

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $-7 + 3$

ANSWER -4

2. $-5 - 4$

ANSWER -9

3. $(-6)(-4)$

ANSWER 24

4. Simplify $3x + 5x - 2x$

ANSWER $6x$

5. Solve $3x + 7 = 22$

ANSWER $x = 5$

6. $\sqrt{144}$

ANSWER 12

7. 3.6×10^3

ANSWER 3600 **Medium · the standard question**

7 PROBLEMS

1. $3 - (-8)$

ANSWER 11

2. $\frac{2}{3} \div \frac{4}{9}$

ANSWER $1\frac{1}{2}$

3. Expand $2(3x - 4)$

ANSWER $6x - 8$

4. Solve $5x - 4 = 3x + 10$

ANSWER $x = 7$

5. $HCF(12, 18)$

ANSWER 6

6. $LCM(12, 18)$

ANSWER 36

7. $6.3 \div 0.7$

ANSWER 9

Hard · stretch and reason

7 PROBLEMS

1. $2\frac{3}{4} + 1\frac{5}{6}$

ANSWER $4\frac{7}{12}$

2. $4\frac{1}{3} - 1\frac{5}{6}$

ANSWER $2\frac{1}{2}$

3. Solve $2(x - 3) = 14$

ANSWER $x = 10$

4. The n th term of 5, 8, 11, 14

ANSWER $3n + 2$

5. The 100th term of $4n - 1$

ANSWER 399

6. 3.847 to 2 d.p.

ANSWER 3.85

7. 0.8×0.04

ANSWER 0.032

07 Homework

Homework — 5 tasks

Show the working line by line; the marks in the annual paper are for the method as much as the answer.

- Work out $-9 + 4 - (-6)$ and $(-8)(-3) \div 4$.
- Work out $3\frac{1}{2} + 2\frac{2}{3}$ and $5\frac{1}{4} - 2\frac{2}{3}$.
- Solve $4x - 5 = 2x + 11$ and check your answer.

4. Find the n th term of $4, 11, 18, 25, \dots$ and its 50th term.
5. Work out 0.9×0.06 , $4.8 \div 0.6$ and 2.5×10^3 .

General revision

II — ratio, percentage and rates

Sharing, proportion, the three percentage questions, percentage change and speed.

LESSON 195–198

4 × 40 MIN

MATEMATIKA 6, YAKUNIY TAKRORLASH

S7 WHOLE-COURSE REVIEW

LESSON CLOCK

30'

35'

35'

35'

20'

5

Ratio	30 min
Sharing and proportion	35 min
The three percentage questions	35 min
Percentage change	35 min
Speed and motion	20 min
Homework	5 min

LEARNING OBJECTIVES

- Simplify a ratio and share a quantity in a given ratio.
- Use direct and inverse proportion correctly.
- Answer all three kinds of percentage question, including the reverse one.
- Use $s = vt$ and find an average speed.

TEXTBOOK

National	pp. 557–568
Cambridge	Stage 7, pp. 120–132
Workbook	Checkpoint practice 2

01

Explanation

Ratio

RATIO	DIVIDE BY	SIMPLEST FORM
12 : 18	6	2 : 3
45 : 60	15	3 : 4
0.5 : 2	multiply by 2 first	1 : 4
$\frac{3}{4} : \frac{1}{2}$	multiply by 4 first	3 : 2

A RATIO COMPARES PARTS; A FRACTION COMPARES A PART WITH THE WHOLE

In $2 : 3$ the first quantity is $\frac{2}{5}$ of the total, not $\frac{2}{3}$. Adding the parts before writing any fraction removes the commonest error of the whole topic.

Sharing and proportion

TASK	PARTS	ONE PART	ANSWER
share 60 in $2 : 3$	5	12	24 and 36
share 84 in $3 : 4$	7	12	36 and 48
share 120 in $2 : 3 : 5$	10	12	24, 36 and 60

KIND	SITUATION	METHOD	ANSWER
direct	5 pens cost 20 000 so'm; 8 pens?	one pen is 4000	32 000 so'm
inverse	4 workers take 12 days; 6 workers?	the job is 48 worker-days	8 days

MORE WORKERS, FEWER DAYS

In direct proportion both quantities rise together; in inverse proportion one rises as the other falls, and the *product* stays the same. Ask which of the two the situation is before calculating.

The three percentage questions

QUESTION	EXAMPLE	METHOD	ANSWER
find a percentage of a number	15% of 240	$0.15 \cdot 240$	36
express one number as a percentage of another	30 out of 120	$\frac{30}{120} \cdot 100$	25%
find the whole from a percentage	20% of it is 45	$45 \div 0.2$	225

NAME THE QUESTION BEFORE CHOOSING THE METHOD

All three use the same three numbers in different places. Writing “this is the third kind” at the top of the working is worth more than remembering three formulas.

Percentage change

$$\text{multiplier} = 1 + \frac{\text{percentage}}{100} \text{ for a rise, } 1 - \frac{\text{percentage}}{100} \text{ for a fall}$$

TASK	MULTIPLIER	WORKING	ANSWER
increase 400 by 15%	1.15	$400 \cdot 1.15$	460
decrease 500 by 20%	0.8	$500 \cdot 0.8$	400
200 rises to 250	—	$\frac{50}{200} \cdot 100$	a 25% rise
80 falls to 60	—	$\frac{20}{80} \cdot 100$	a 25% fall
after a 20% rise a price is 60 000	1.2	$60\,000 \div 1.2$	50 000 so'm

THE CHANGE IS ALWAYS MEASURED AGAINST THE ORIGINAL

200 to 250 is a rise of 25%, but 250 back to 200 is a fall of 20%. The same difference, two different percentages, because the starting numbers differ.

Speed and motion

$$s = vt \quad v = \frac{s}{t} \quad t = \frac{s}{v}$$

QUESTION	WORKING	ANSWER
180 km in 3 h	$180 \div 3$	60 km/h
4 km in 20 min	$4 \div \frac{1}{3}$	12 km/h
7 km at 5 km/h	1.4 h	1 h 24 min
72 km/h in m/s	$72 \div 3.6$	20 m/s
120 km at 60, back at 40	$240 \div 5$	48 km/h
two cars 300 km apart at 60 and 40	$300 \div 100$	meet after 3 h

UNITS FIRST, FORMULA SECOND

Minutes become hours before they meet a speed in km/h; an average speed is total distance over total time and never the mean of the speeds.

02 Worked examples

EXAMPLE 1 Share 120 in the ratio 2 : 3 : 5.

① $2 + 3 + 5 = 10$ parts.

② $120 \div 10 = 12$ for one part.

③ 24, 36 and 60.

They add to 120 ✓

ANSWER

24, 36 and 60

EXAMPLE 2 After a 20% rise a price is 60 000 so'm. Find the original price.

① The multiplier for a 20% rise is 1.2.

② The original times 1.2 is 60 000.

③ $60\,000 \div 1.2 = 50\,000$ so'm.

Check: $50\,000 \cdot 1.2 = 60\,000$ ✓

ANSWER

50 000 so'm

EXAMPLE 3 A car drives 120 km at 60 km/h and returns at 40. Find the average speed.

- ① Out 2 h, back 3 h.
- ② Total 240 km in 5 h.
- ③ 48 km/h.
Not 50 — more time at the slower speed.

ANSWER

48 km/h

03 Interactive model

Put the three percentage questions on the board with the same three numbers in each and let the class sort them; naming the question is the skill being revised.

Ratio, percentage and rates in eight questions

12 : 18 in simplest form:

2 : 3

3 : 2

6 : 9

4 : 6

Sharing 60 in 2 : 3 gives:

20 and 40

24 and 36

30 and 30

12 and 18

15% of 240 is:

16

36

24

3.6

30 as a percentage of 120 is:

4%

25%

30%

40%

Increasing 400 by 15% gives:

415

460

340

600

After a 20% rise a price is 60 000; it was:

48 000

50 000

54 000

72 000

4 km in 20 minutes is:

0.2

12

80

4

4 workers take 12 days; 6 workers take:

18

8

6

12

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Ratio	Nisbat	Отношение
To share in a ratio	Nisbatda taqsimlash	Делить в отношении
Direct proportion	To'g'ri proporsionallik	Прямая пропорциональность
Inverse proportion	Teskari proporsionallik	Обратная пропорциональность
Percentage	Foiz	Процент
Percentage change	Foizli o'zgarish	Процентное изменение
Multiplier	Ko'paytuvchi	Множитель
Average speed	O'rtacha tezlik	Средняя скорость

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

In the ratio 2 : 3 the first part is:

$\frac{2}{3}$ of the total

$\frac{2}{5}$ of the total

half

$\frac{3}{5}$ of the total

More workers taking fewer days is:

direct proportion

inverse proportion

a ratio

a percentage

The multiplier for a 20% fall is:

0.2

0.8

1.2

20

A percentage change is measured against:

the new value

the original value

the difference

a hundred

Average speed is:

the mean of the speeds

total distance over total time

the largest speed

the first speed

72 km/h in m/s is:

7.2

20

26

259

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Simplify $12 : 18$

ANSWER $2 : 3$

2. Simplify $45 : 60$

ANSWER $3 : 4$

3. Share 60 in the ratio $2 : 3$

ANSWER 24 and 36

4. 15% of 240

ANSWER 36

5. 180 km in 3 hours

ANSWER 60 km/h

6. 30 as a percentage of 120

ANSWER 25%

7. 5 pens cost 20 000 so'm: one pen

ANSWER 4000 so'm

Medium · the standard question

7 PROBLEMS

1. Share 84 in the ratio 3 : 4

ANSWER 36 and 48

2. Share 120 in the ratio 2 : 3 : 5

ANSWER 24, 36 and 60

3. 8 pens at 4000 so'm each

ANSWER 32 000 so'm

4. Increase 400 by 15%

ANSWER 460

5. Decrease 500 by 20%

ANSWER 400

6. 4 km in 20 minutes

ANSWER 12 km/h

7. 72 km/h in m/s

ANSWER 20 m/s

Hard · stretch and reason

7 PROBLEMS

1. Simplify $\frac{3}{4} : \frac{1}{2}$

ANSWER 3 : 2

2. 4 workers take 12 days: 6 workers

ANSWER 8 days

3. 200 rising to 250

ANSWER A 25% increase

4. 80 falling to 60

ANSWER A 25% decrease

5. After a 20% rise a price is 60 000 so'm: the original

ANSWER 50 000 so'm

6. 120 km at 60 km/h and back at 40

ANSWER 48 km/h

7. Two cars 300 km apart at 60 and 40 km/h meet after

ANSWER 3 hours

07 Homework

Homework — 5 tasks

Name the kind of question at the top of each solution before working it.

- Share 150 in the ratio 3 : 7 and 180 in the ratio 1 : 2 : 3.
- Find 18% of 350, and express 42 as a percentage of 120.
- A price falls by 15% to 68 000 so'm. Find the original price.
- 6 workers take 10 days. How long would 4 workers take?

5. A cyclist rides 30 km at 15 km/h and 30 km at 10. Find the average speed.

General revision

III — geometry, mensuration and data

Angles, the circle, areas, solids, and the whole of the data and probability block.

LESSON 199–202

4 × 40 MIN

MATEMATIKA 6, YAKUNIY TAKRORLASH

S7 WHOLE-COURSE REVIEW

LESSON CLOCK

30'

35'

35'

35'

20'

5

Angles	30 min
The circle	35 min
Area and compound figures	35 min
Solids	35 min
Data and probability	20 min
Homework	5 min

LEARNING OBJECTIVES

- Find unknown angles at a point, on a line and inside a polygon.
- Find the circumference and the area of a circle and its parts.
- Find areas of compound figures and volumes of prisms.
- Find averages, sector angles and probabilities.

TEXTBOOK

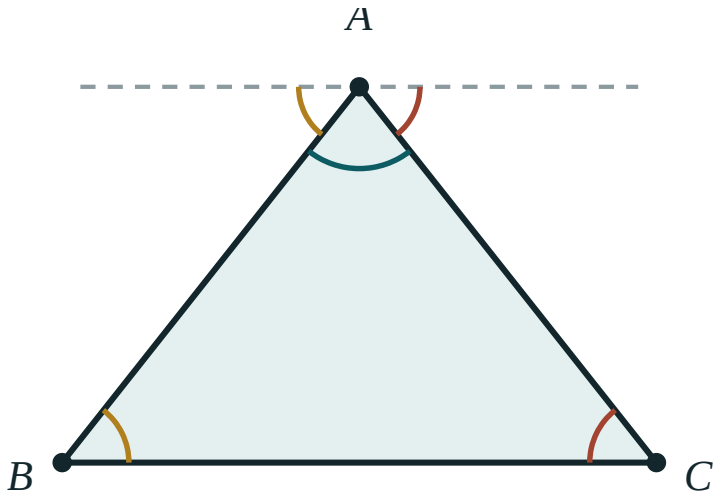
National	pp. 569–580
Cambridge	Stage 7, pp. 40–188
Workbook	Checkpoint practice 3

01

Explanation

Angles

FACT	TOTAL	EXAMPLE	UNKNOWN
angles on a straight line	180°	55° and x	125°
angles at a point	360°	90°, 130° and x	140°
angles of a triangle	180°	40°, 65° and x	75°
angles of a quadrilateral	360°	100°, 85°, 95° and x	80°
vertically opposite angles	equal	x opposite 62°	62°



the three angles fill 180°

The three angles of a triangle make a straight line

AN ISOSCELES TRIANGLE GIVES TWO FACTS AT ONCE

Two equal sides mean two equal angles. With an apex of 40° the other two share 140°, so each is 70°.

The circle

$C = 2\pi r = \pi d$ and $A = \pi r^2$

CIRCLE	CIRCUMFERENCE	AREA
$r = 5\text{ cm}$	31.4 cm	78.5 cm^2
$r = 10\text{ cm}$	62.8 cm	314 cm^2
$d = 14\text{ cm}$	about 44 cm	about 154 cm^2

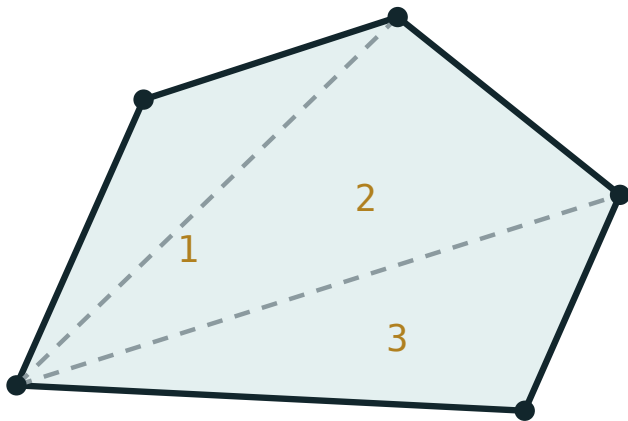
PART OF A CIRCLE	WORKING	ANSWER
perimeter of a semicircle, $r = 5$	$15.7 + 10$	25.7 cm
area of a semicircle, $r = 6$	$113.04 \div 2$	56.52 cm^2
area of a quarter circle, $r = 4$	$50.24 \div 4$	12.56 cm^2

THE PERIMETER OF A SEMICIRCLE INCLUDES THE DIAMETER

Half the circumference is only the curve. The straight edge across the bottom is part of the boundary and must be added — $15.7 + 10$, not 15.7 .

Area and compound figures

FIGURE	RULE	EXAMPLE	AREA
rectangle	ab	12×7	84 cm^2
triangle	$\frac{1}{2}bh$	base 10, height 6	30 cm^2
parallelogram	bh	8×5	40 cm^2
trapezium	$\frac{a + b}{2} \cdot h$	6 and 10, height 4	32 cm^2
an L-shape	subtract the missing corner	$10 \times 6 - 4 \times 3$	48 cm^2



split into triangles, then add the areas

A compound figure cut into pieces whose areas are already known

CUT IT UP, OR FILL IT IN

Every compound figure is either a sum of simple pieces or a large shape with pieces removed. Draw the cutting lines on the diagram before writing any numbers.

Solids

SOLID	VOLUME	SURFACE AREA
$5 \times 4 \times 3\text{ cm cuboid}$	60 cm^3	94 cm^2
cube of edge 4 cm	64 cm^3	96 cm^2
prism, cross-section 12 cm^2 , length 10 cm	120 cm^3	—
cylinder $r = 5, h = 10$	785 cm^3	471 cm^2

SOLID	FACES	EDGES	VERTICES
cube	6	12	8
triangular prism	5	9	6
square pyramid	5	8	5

ONE RULE COVERS EVERY PRISM

Volume is the cross-section area times the length, whether the end is a rectangle, a triangle or a circle.
Surface area is the total of the pieces of the net.

Data and probability

TASK	EXAMPLE	ANSWER
mean	2, 5, 5, 6, 12	6
median	the same set	5
mode and range	the same set	5 and 10
sector angle	12 out of 36	120°
reading a sector	90° of 200 people	50 people
probability	$P(\text{even})$ on one throw of a die	$\frac{1}{2}$
the complement	$P(\text{not red})$ with 4 red in 10	$\frac{3}{5}$

A SECTOR ANGLE NEEDS THE TOTAL; A PROBABILITY NEEDS THE WHOLE SET

Both are shares. The commonest lost mark in this block is using the frequency where the total was needed, or the other way round.

02

Worked examples

EXAMPLE 1 A circle has radius 6 cm. Find the area of a semicircle on it, taking $\pi = 3.14$.

① $A = 3.14 \cdot 36 = 113.04 \text{ cm}^2$.

② Half of that.

③ 56.52 cm^2 .

Taking $\pi = 3.14$ ✓

ANSWER

56.52 cm^2

EXAMPLE 2 An L-shape is a 10 by 6 cm rectangle with a 4 by 3 cm corner removed. Find its area.

① $10 \cdot 6 = 60 \text{ cm}^2$.

② $4 \cdot 3 = 12 \text{ cm}^2$ removed.

③ $60 - 12 = 48 \text{ cm}^2$.

Draw the cut before calculating ✓

ANSWER

48 cm^2

EXAMPLE 3 A pie chart of 200 people has a 90° sector. How many people, and what percentage?

① 90° is a quarter of the circle.

② $200 \div 4 = 50$ people.

③ $90 \div 3.6 = 25\%$.
A quarter either way ✓

ANSWER

50 people, or 25%

03 Interactive model

Set one question from each of the five sections as a timed round; the switching between topics is the part of an annual paper that catches people out.

Geometry, mensuration and data in eight questions

Angles at a point total:

90°

180°

270°

360°

40° , 65° and x in a triangle gives:

65°

75°

85°

105°

The circumference of a circle of radius 5 cm is:

15.7

31.4

78.5

10

Its area is:

15.7

31.4

78.5

25

The perimeter of a semicircle of radius 5 cm is:

15.7

25.7

31.4

20

A trapezium with parallel sides 6 and 10 and height 4 has area:

24

32

40

64

A cylinder $r = 5$, $h = 10$ holds:

157

314

471

785

$P(\text{not red})$ with 4 red in 10 is:

$$\frac{2}{5}$$

$$\frac{3}{5}$$

$$\frac{1}{2}$$

$$\frac{6}{4}$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Angles on a line	Yoyiq burchak	Развёрнутый угол
Angles at a point	To'la burchak	Полный угол
Circumference	Aylana uzunligi	Длина окружности
Area of a circle	Doira yuzasi	Площадь круга
Compound figure	Murakkab shakl	Составная фигура
Prism	Prizma	Призма
Sector angle	Sektor burchagi	Угол сектора
Probability	Ehtimollik	Вероятность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Vertically opposite angles are:

supplementary

equal

complementary

unrelated

The area of a circle is:

$2\pi r$

πr^2

πd

$2\pi r^2$

A compound figure is handled by:

guessing

cutting it into known pieces

measuring

ignoring part of it

The volume of any prism is:

base area $\times$ length

all edges multiplied

surface $\times$ 3

base $\div$ height

A sector angle is found from:

the frequency alone

the frequency and the total

the total alone

the mean

A probability of 1.4 means:

a likely event

a mistake

a certain event

a percentage

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Angles on a straight line total

ANSWER 180°

2. Angles at a point total

ANSWER 360°

3. The angles of a triangle total

ANSWER 180°

4. The angles of a quadrilateral total

ANSWER 360°

5. The circumference of a circle of radius 5 cm

ANSWER 31.4 cm

6. The area of a rectangle 12 by 7 cm

ANSWER 84 cm^2

7. The mode of 2, 5, 5, 6, 12

ANSWER 5

Medium · the standard question

7 PROBLEMS

1. 55° and x on a straight line

ANSWER 125°

2. 90° , 130° and x at a point

ANSWER 140°

3. 40° , 65° and x in a triangle

ANSWER 75°

4. The area of a circle of radius 5 cm

ANSWER 78.5 cm^2

5. The area of a triangle, base 10, height 6

ANSWER 30 cm^2

6. The area of a trapezium, sides 6 and 10, height 4

ANSWER 32 cm^2

7. 12 out of 36 as a sector angle

ANSWER 120°

Hard · stretch and reason

7 PROBLEMS

1. The base angles of an isosceles triangle with apex 40°

ANSWER 70° each

2. The circumference of a circle of diameter 14 cm

ANSWER About 44 cm

3. The perimeter of a semicircle of radius 5 cm

ANSWER 25.7 cm

4. The area of a semicircle of radius 6 cm

ANSWER 56.52 cm^2

5. The volume and surface of a cylinder $r = 5$, $h = 10$

ANSWER 785 cm^3 and 471 cm^2

6. 200 people, a sector of 90°

ANSWER 50 people

7. $P(\text{not red})$ with 4 red in 10

ANSWER $\frac{3}{5}$

07 Homework

Homework — 5 tasks

Write the geometric fact or the formula beside every line of working.

- Find x when x , 70° and 45° are the angles of a triangle, and again when they are angles at a point with 115° .
- Find the circumference and the area of a circle of radius 7 cm, taking $\pi = 3.14$.
- Find the area of an L-shape made from a 12 by 8 cm rectangle with a 5 by 4 cm corner removed.
- Find the volume and the surface area of a cuboid 9 cm by 4 cm by 3 cm.

5. A pie chart of 45 pupils has a group of 15. Find the sector angle, and find $P(\text{odd})$ for one throw of a die.

Presentation and assessment of the project work

The survey, the pie chart and the conclusions — presented, questioned and marked.

LESSON 203–204

2 × 40 MIN

MATEMATIKA 6, LOYIHA HIMOYASI

S7 PROJECT PRESENTATION

LESSON CLOCK

10'

25'

25'

20'

What is presented	10 min
The presentations	25 min
The assessment criteria	25 min
Questions, and closing the year	20 min

LEARNING OBJECTIVES

- Present a survey, its chart and its conclusions in four minutes.
- Justify the sample and state one honest limit of the result.
- Answer questions about the data with the numbers in front of you.
- Assess another project against the published criteria.

TEXTBOOK

National	pp. 581–584
Cambridge	Stage 7, pp. 184–188
Workbook	Project assessment

01

Explanation

What is presented

Each pupil brings one poster and speaks for four minutes. The mathematics of the project is everything from lessons 157 to 168, applied to data they collected themselves.

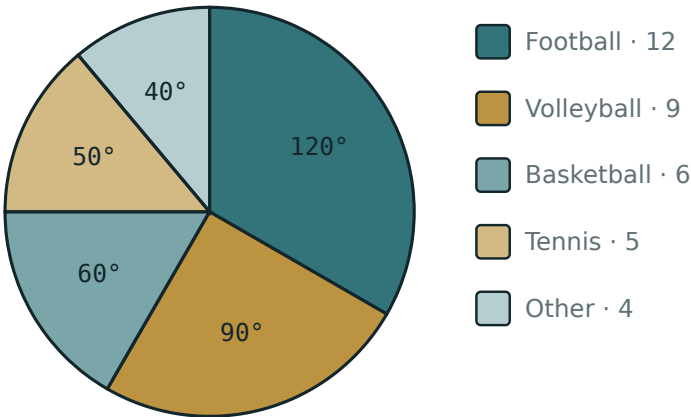
ON THE POSTER	FROM
the question, with its list of answers	L158–159
the sample: who was asked, how many, and how they were chosen	L158–159
the frequency table, with the total	L157
the pie chart, with every sector angle written on	L160–163
three conclusions and one honest limit	L164–168

THE TOTAL IS NOT DECORATION

Without the number surveyed written on the poster, no sector can be turned back into people. It is the first thing the audience will ask for.

The presentations

A worked example, from a survey of 60 pupils.



SPORT	PUPILS	ANGLE	PERCENTAGE
football	20	120°	33.3%
volleyball	15	90°	25%
basketball	10	60°	16.7%
tennis	9	54°	15%
swimming	6	36°	10%
total	60	360°	100%

ONE PUPIL IS 6°

360 ÷ 60. Every angle in the table is the frequency times six, and the check is that the column adds to 360°.

The assessment criteria

CRITERION	WHAT EARNS THE MARKS	MARKS
the question and the sample	fixed options, no gaps or overlaps; the sample justified	4
the data and the table	at least 60 asked; frequencies adding to the total	4
the pie chart	angles correct and adding to 360° ; labels or a key; the total shown	5
the conclusions	three statements from the data, and one honest limit	4
the talk	four minutes, clear, and questions answered	3
total	—	20

A BEAUTIFUL CHART WITH THE WRONG ANGLES SCORES BADLY

Five of the twenty marks are for the chart, and most of those are for the arithmetic behind it. Check the angle column against 360° before the poster is glued.

Questions, and closing the year

QUESTION THE AUDIENCE MAY ASK	WHAT A GOOD ANSWER USES
how many people is that sector?	the angle and the total
what percentage is it?	the angle divided by 3.6
would the whole school agree?	the sample, and its honest limit
if the school has 600 pupils, how many play football?	the share, scaled up: 200
why a pie chart and not a bar chart?	the data is shares of one whole

THE YEAR, IN ONE LINE

Directed numbers and fractions; expressions and equations; ratio, proportion and percentage; the circle and the solids; speed; and data with probability. Every one of them appeared in somebody's project.

02 Worked examples

EXAMPLE 1 A survey of 60 pupils gives 20, 15, 10, 9 and 6. Find the five sector angles.

- ① $360 \div 60 = 6^\circ$ for each pupil.
- ② Multiply each frequency by 6.
- ③ $120^\circ, 90^\circ, 60^\circ, 54^\circ$ and 36° .
They total 360° ✓

ANSWER

$120^\circ, 90^\circ, 60^\circ, 54^\circ$ and 36°

EXAMPLE 2 The school has 600 pupils. Estimate how many prefer football, and say what the estimate assumes.

- ① Football is $\frac{20}{60} = \frac{1}{3}$ of the sample.
- ② $\frac{1}{3} \cdot 600 = 200$ pupils.
- ③ It assumes the sample is representative of the school.
That is the honest limit ✓

ANSWER

About 200, if the sample represents the school

EXAMPLE
3**Two more pupils are asked and both choose football. Find the new sector angle, to 1 decimal place.**

① Now 22 of 62.

② $\frac{22}{62} \cdot 360.$

③ $= 127.7^\circ.$

Every other angle changes too — the total is still 360° .**ANSWER** 127.7° **03 Interactive model**

Give each pupil the criteria table and one other project to mark; explaining a lost mark to a classmate teaches the criterion better than losing it yourself.

Reading a classmate's project

With 60 surveyed, one pupil is worth:

20 of the 60 gives a sector of:

That is what percentage?

A 54° sector of 60 pupils is:

6

9

15

54

The angle column must total:

100

180°

360°

the number surveyed

Scaling 1 3 up to a school of 600 gives:

60

180

200

300

A poster with no total:

is complete

cannot be turned into numbers

is more accurate

scores full marks

Most of the chart marks are for:

the colours

the arithmetic behind the angles

the size

the handwriting

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Presentation	Taqdimot	Презентация
Poster	Plakat	Плакат
Conclusion	Xulosa	Вывод
Criterion	Mezon	Критерий
Peer assessment	O'zaro baholash	Взаимооценивание
To justify	Asoslamog	Обосновать
Limitation	Cheklov	Ограничение
Scaling up	Kengaytirib baholash	Перенос на всю совокупность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The first thing an audience asks for is:

the colours

the number surveyed

the title

the date

Three conclusions should come from:

opinion

the data

the textbook

the teacher

An honest limit is:

a weakness to hide

part of a good report

a lost mark

optional

A pie chart suits this project because the data is:

ordered in time

shares of one whole

two measurements

continuous

Scaling a share up to the whole school assumes:

nothing

the sample is representative

the school is small

the chart is neat

Peer assessment is done against:

a feeling

the published criteria

the largest poster

the loudest talk

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The total surveyed in the example

ANSWER 60

2. The angle representing one pupil

ANSWER 6°

3. The sector for football, 20 pupils

ANSWER 120°

4. The sector for volleyball, 15 pupils

ANSWER 90°

5. The sector for basketball, 10 pupils

ANSWER 60°

6. The angles of the chart total

ANSWER 360°

7. The largest group in the example

ANSWER Football

Medium · the standard question

7 PROBLEMS

1. The sector for tennis, 9 pupils

ANSWER 54°

2. The sector for swimming, 6 pupils

ANSWER 36°

3. Football as a percentage

ANSWER 33.3%

4. Volleyball as a percentage

ANSWER 25%

5. Swimming as a percentage

ANSWER 10%

6. How many chose a ball game

ANSWER 45 pupils

7. ...as a sector angle

ANSWER 270°

Hard · stretch and reason

7 PROBLEMS

1. Basketball as a percentage, to 1 d.p.

ANSWER 16.7%

2. If the school of 600 followed this pattern, the footballers

ANSWER 200

3. One honest limit of this survey

ANSWER Only 60 of 600 pupils were asked

4. Why must the angle column be checked against 360° ?

ANSWER Rounding can leave the chart open

5. Two more pupils choose football: the new sector, to 1 d.p.

ANSWER 127.7°

6. The better chart for comparing the sizes of six grades

ANSWER A bar chart

7. The three things every chart must carry

ANSWER A title, labels or a key, and the total

07 Homework

Homework — the reflection

Half a page, handed in with the poster; it is read, not marked out of twenty.

1. Mark your own project against the five criteria and give yourself a total out of 20.
2. Write the one criterion you would most like to improve, and how.
3. Write the question from the audience that was hardest to answer, and your answer to it now.
4. Name the topic of the year you found most useful outside mathematics, and why.
5. Write one question about Grade 7 mathematics that you would like answered in September.